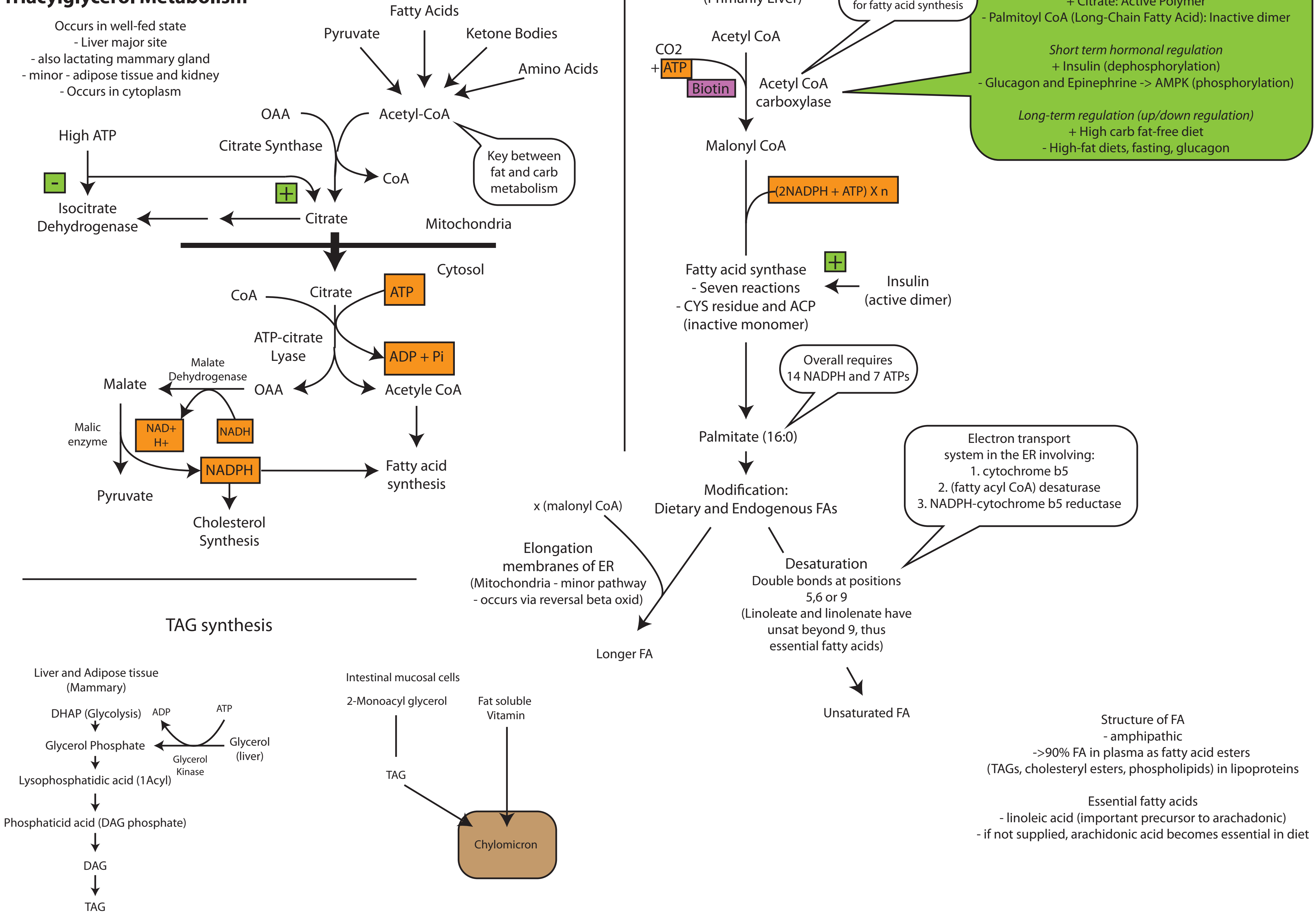
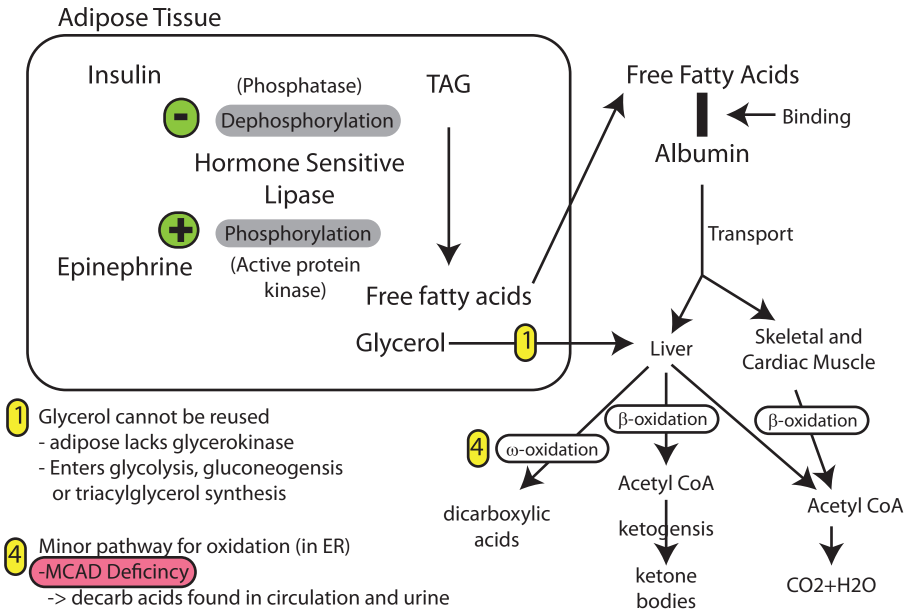
Fatty Acid and Triacylglycerol Metabolism

Occurs in well-fed state
- Liver major site
- also lactating mammary gland
- minor - adipose tissue and kidney
- Occurs in cytoplasm



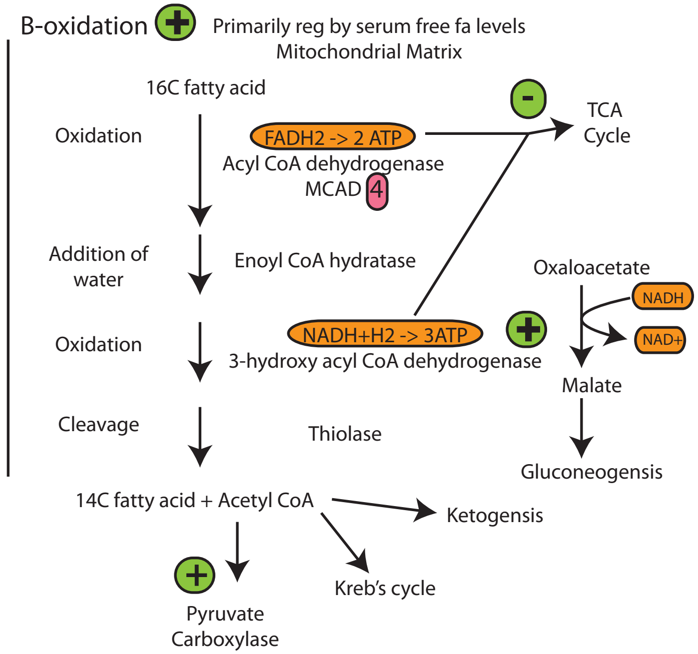
Structure of FA
- amphipathic
->90% FA in plasma as fatty acid esters
(TAGs, cholesteryl esters, phospholipids) in lipoproteins

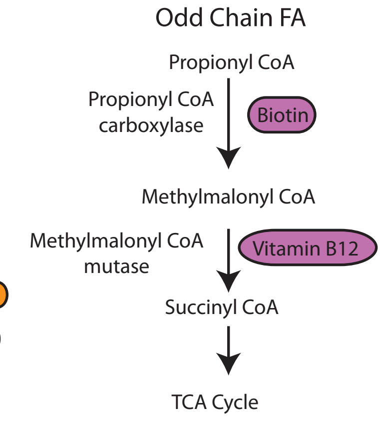
Essential fatty acids
- linoleic acid (important precursor to arachadonic)
- if not supplied, arachidonic acid becomes essential in diet

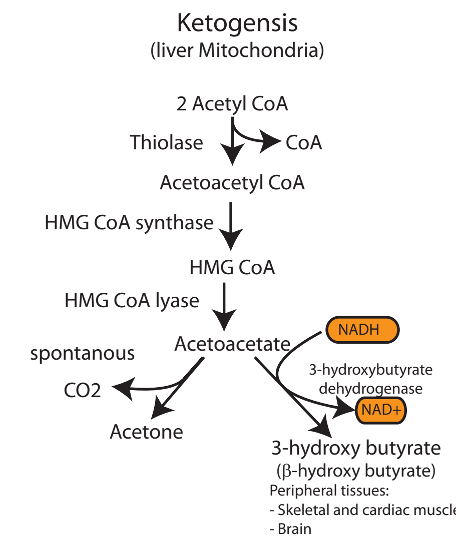


1 Glycerol cannot be reused
- adipose lacks glycerokinase
- Enters glycolysis, gluconeogensis or triacylglycerol synthesis

4 Minor pathway for oxidation (in ER)
-MCAD Deficincy
-> decarb acids found in circulation and urine

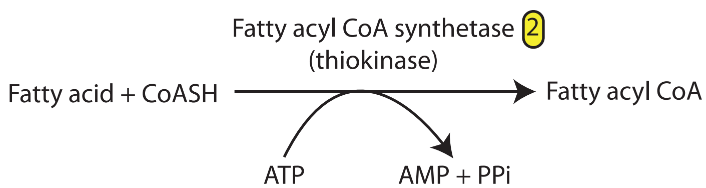


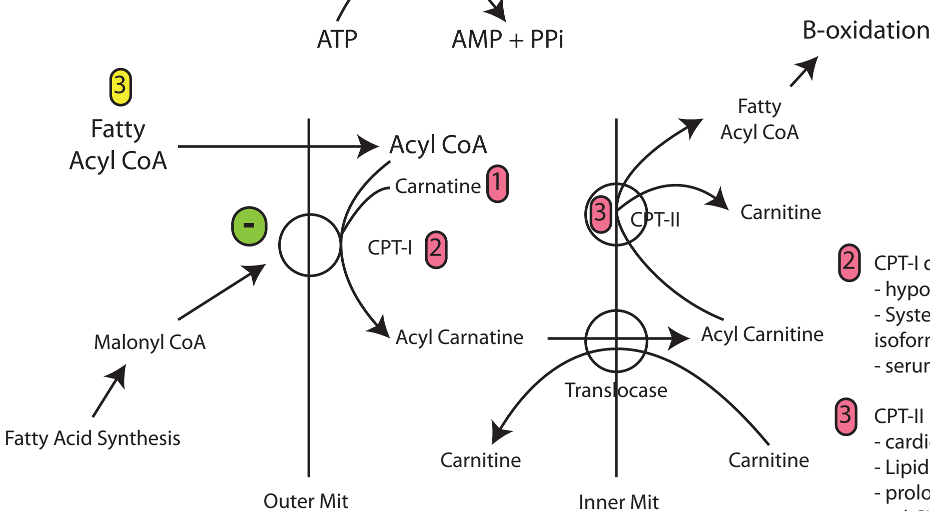




4 MCAD deficiency
- Most common autosomal recessive enzyme def
- Decreased ability to oxidize 6-10C FA
- Hypoglycemia during fasting (liver and muscle not utilize FA) due to high glycolysis
- Medium chain (8-10C) acyl carnitines excreted in urine
-Dicarboxylic acids found in urine (high omega oxid)
- CK-MM and myglobin

JAMAICAN VOMITING SICKNESS - unripe ackee fruit
- hypoglycin A inhibits MCAD (beta oxidation)
- vomiting after ingest, drowsiness (hypoglyc), coma and death



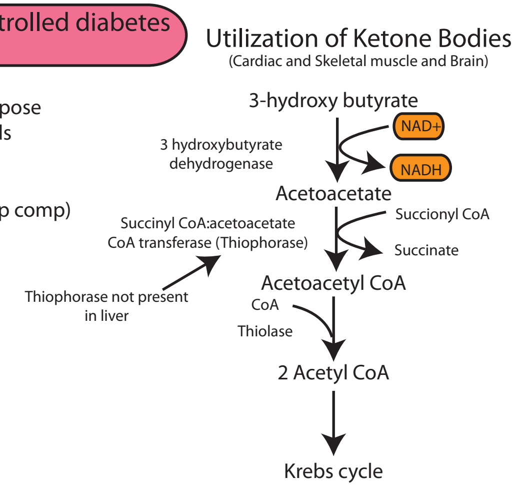


2 CPT-I deficiency
- hypoglycemia and hypoketonemia
- Systemic form: Predominantly affects liver isoform
- serum carnitine levels usually elevated

3 CPT-II Deficiency
- cardiomyopathy and myopathic
- Lipid deposits in skeletal muscle
- prolonged exercise -> myoglobinuria and CK-MM levels in serum

Ketoacidosis in uncontrolled diabetes mellitus

- Very low insulin
- Super active lipoysis in adipose
- Ketone bodies = weak acids
- HCO3- buffer thus acidosis
- Ketone bodies in urine
- Resipiration increased (resp comp)



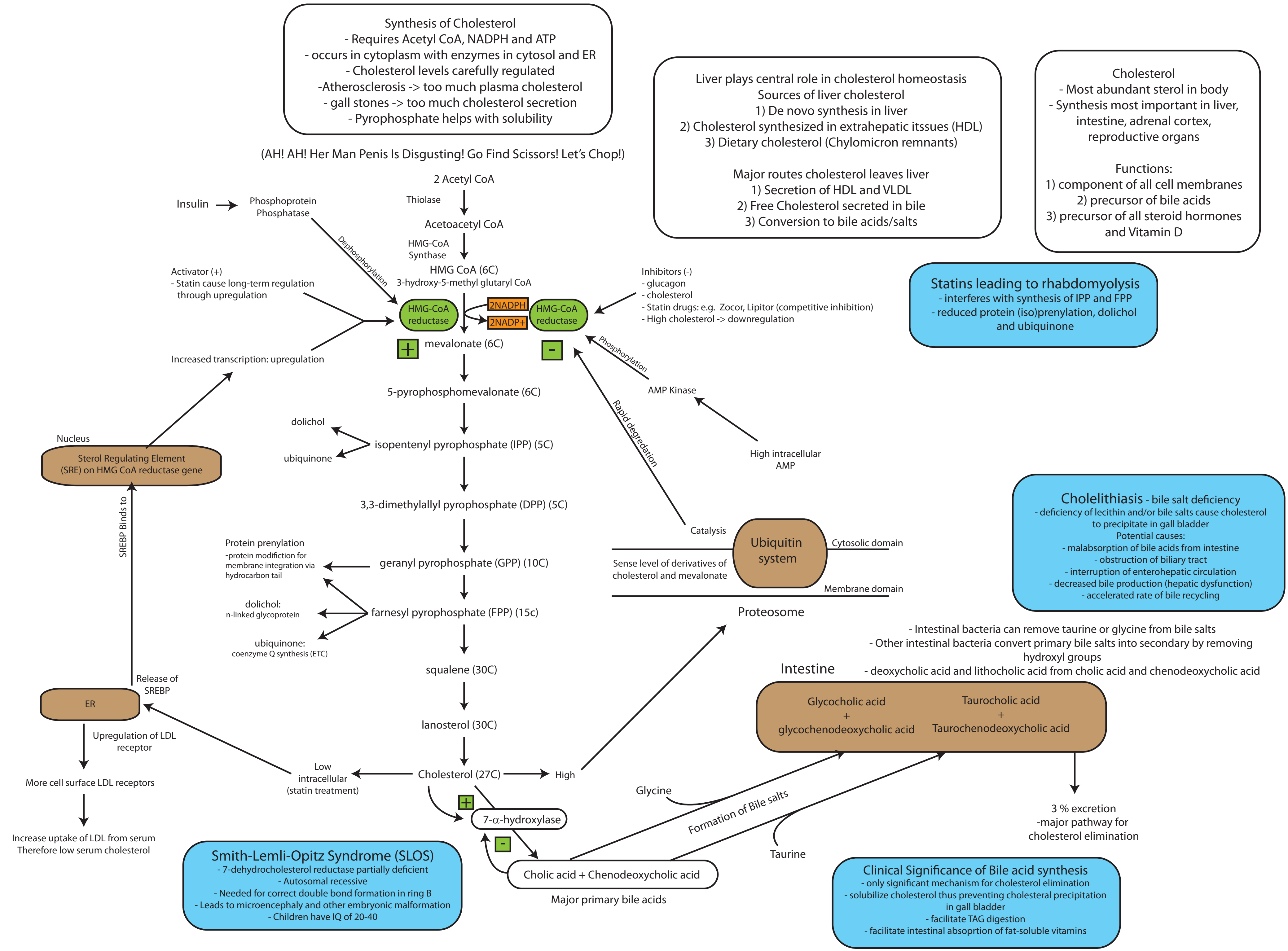
2 Present in outer mitochondrial membrane (cytosol side)

3 Fatty acid chains shorter than shorter than 12 C atoms can cross without carnitine or CPT

1 Primary Carnitine deficiency
- carnitine uptake into tissues impared
- transport of long chain FA into Mit impaired
- B oxidation decreased
- Hypoglycemia and hypoketonemia
-> impaired gluconeogensis (pyruvate carboxy)
- Systemic car def - ketogensis decreased if liver carnitine deficient. Presents at early age
- Reduced serum carnitine
*Secondary-> liver disease/chronic renal failure

Increased Ketogensis during Starvation

1) decreased insulin glucagon ratio, activation of hormone sensitive lipase
2) Inc FFA = Inc B-oxid due to absence inhibit CPT-I by malonyl CoA
3) Inc B-ox = inc NADH/nAD+ and inc ATP
4) Acetyl CoA stimulates Pyruvate Carbox and pyruvate shunted to glyconeogensis and Inc NADH/NAD+ = malate
5) Acetyl Coa shunted towards ketogensis rather than TCA



Synthesis of Cholesterol
- Requires Acetyl CoA, NADPH and ATP
- occurs in cytoplasm with enzymes in cytosol and ER
- Cholesterol levels carefully regulated
-Atherosclerosis -> too much plasma cholesterol
- gall stones -> too much cholesterol secretion
- Pyrophosphate helps with solubility
(AH! AH! Her Man Penis Is Disgusting! Go Find Scissors! Let's Chop!)
Liver plays central role in cholesterol homeostasis
Sources of liver cholesterol
1) De novo synthesis in liver
2) Cholesterol synthesized in extrahepatic itssues (HDL)
3) Dietary cholesterol (Chylomicron remnants)
Major routes cholesterol leaves liver
1) Secretion of HDL and VLDL
2) Free Cholesterol secreted in bile
3) Conversion to bile acids/salts
Cholesterol
- Most abundant sterol in body
- Synthesis most important in liver, intestine, adrenal cortex, reproductive organs
Functions:
1) component of all cell membranes
2) precursor of bile acids
3) precursor of all steroid hormones and Vitamin D
2 Acetyl CoA
Thiolase
Acetoacetyl CoA
HMG-CoA Synthase
HMG CoA (6C)
3-hydroxy-5-methyl glutaryl CoA
Insulin
Phosphoprotein Phosphatase
Dephosphorylation
Activator (+)
- Statin cause long-term regulation through upregulation
Increased transcription: upregulation
HMG-CoA reductase
2NADPH
2NADP+
HMG-CoA reductase
+
-
mevalonate (6C)
Inhibitors (-)
- glucagon
- cholesterol
- Statin drugs: e.g. Zocor, Lipitor (competitive inhibition)
- High cholesterol -> downregulation
Statins leading to rhabdomyolysis
- interferes with synthesis of IPP and FPP
- reduced protein (iso)prenylation, dolichol and ubiquinone
Phosphorylation
AMP Kinase
High intracellular AMP
Rapid degredation
5-pyrophosphomevalonate (6C)
dolichol
ubiquinone
isopentenyl pyrophosphate (IPP) (5C)
Nucleus
Sterol Regulating Element (SRE) on HMG CoA reductase gene
SREBP Binds to
3,3-dimethylallyl pyrophosphate (DPP) (5C)
Catalysis
Ubiquitin system
Cytosolic domain
Sense level of derivatives of cholesterol and mevalonate
Membrane domain
Proteosome
Cholelithiasis - bile salt deficiency
- deficiency of lecithin and/or bile salts cause cholesterol to precipitate in gall bladder
Potential causes:
- malabsorption of bile acids from intestine
- obstruction of biliary tract
- interruption of enterohepatic circulation
- decreased bile production (hepatic dysfunction)
- accelerated rate of bile recycling
Protein prenylation
-protein modifiction for membrane integration via hydrocarbon tail
geranyl pyrophosphate (GPP) (10C)
dolichol:
n-linked glycoprotein
farnesyl pyrophosphate (FPP) (15c)
ubiquinone:
coenzyme Q synthesis (ETC)
squalene (30C)
lanosterol (30C)
- Intestinal bacteria can remove taurine or glycine from bile salts
- Other intestinal bacteria convert primary bile salts into secondary by removing hydroxyl groups
- deoxycholic acid and lithocholic acid from cholic acid and chenodeoxycholic acid
Intestine
Glycocholic acid
+
glycochenodeoxycholic acid
Taurocholic acid
+
Taurochenodeoxycholic acid
Release of SREBP
ER
Upregulation of LDL receptor
More cell surface LDL receptors
Increase uptake of LDL from serum
Therefore low serum cholesterol
Low intracellular (statin treatment)
Cholesterol (27C)
High
+
7-α-hydroxylase
-
Glycine
Formation of Bile salts
Taurine
3 % excretion
-major pathway for cholesterol elimination
Cholic acid + Chenodeoxycholic acid
Major primary bile acids
Smith-Lemli-Opitz Syndrome (SLOS)
- 7-dehydrocholesterol reductase partially deficient
- Autosomal recessive
- Needed for correct double bond formation in ring B
- Leads to microencephaly and other embryonic malformation
- Children have IQ of 20-40
Clinical Significance of Bile acid synthesis
- only significant mechanism for cholesterol elimination
- solubilize cholesterol thus preventing cholesteral precipitation in gall bladder
- facilitate TAG digestion
- facilitate intestinal absoprtion of fat-soluble vitamins

Cholesterol and Steroid Metabolism:

- Cholesterol is precursor of five classes of steroids, glucocorticoids, mineralocorticoids, androgens and progestins
- Synthesis and secretion in:
  1) Adrenal cortex (cortisol, aldosterone)
  2) Ovaries and ovarian corpus luteum (estrogens and progestins)
  3) Testes (testosterone)

Steroid hormones

- use carrier proteins in blood
  1) transcortin for cortisol
  2) sex hormone binding protein for sex hormones
  3) can non-specifically bind albumin

Cholesterol used for steroid hormone synthesis

- taken up via LDL, HDL or synthesized
- stored as cholesterol ester
- PKA activates cholesterol esterase
- transport of cholesterol into mitochondria important regulatory step of cortisol synth
- Performed by StAR (steroidogenic acute regulatory protein)

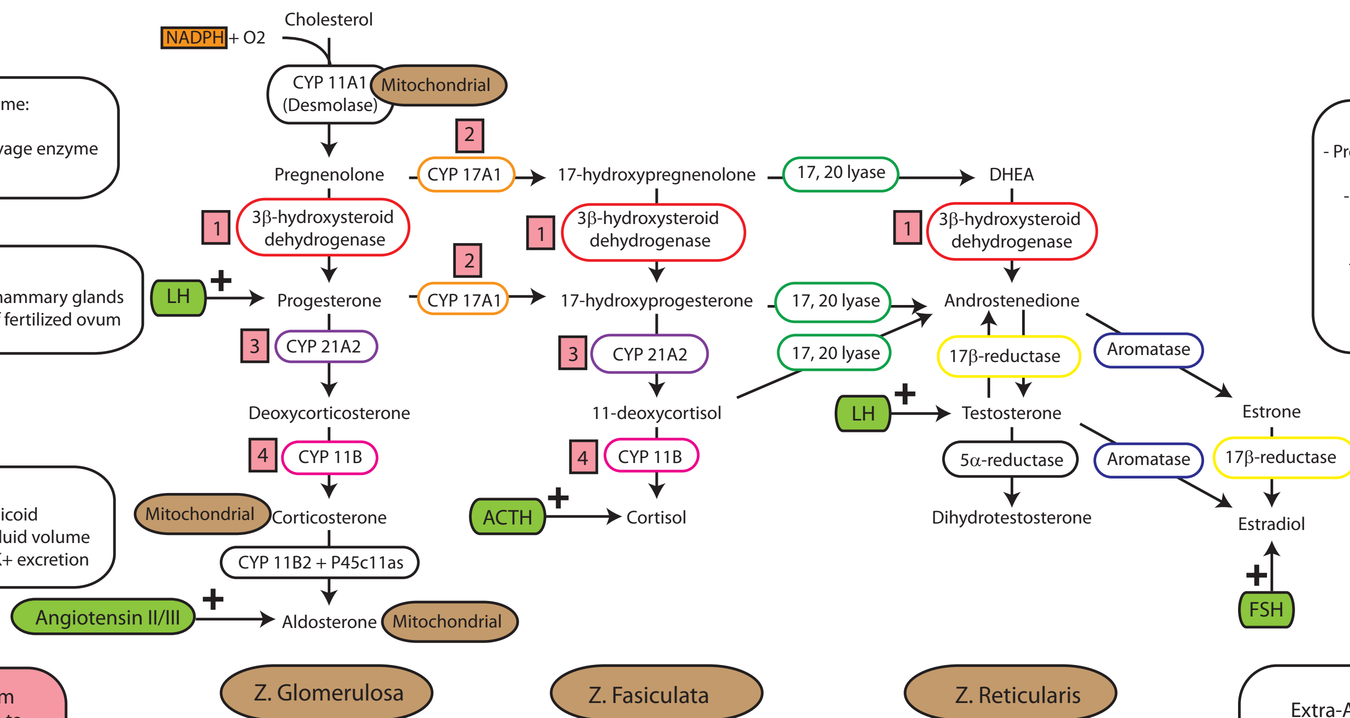
1) 3-β-hydroxysteroid dehydrogenase deficiency
   - Virtually no glucocorticoids, mineralocorticoids, active androgens, or estrogens
   - Marked salt excretion in urine
   - Patients have female-like genitalia

2) 17-α-Hydroxylase Deficiency
   - Virtually no sex hormones or cortisol are produced
   - Increased production of mineralocorticoids cause sodium and fluid retention -> hypertension
   - Patients have female-like genitalia

3) 21-α-Hydroxylase Deficiency
   - Most common form of CAH
   - Mineralocorticoids and glucocorticoids are virtually absent (classic form) or deficient (non-classic)
   - Overproduction of androgens leads to masculinization of external genitalia in females and early virilization in men

4) 11-β-Hydroxylase Deficiency
   - Decrease in serum cortisol, aldosterone, and corticosterone
   - Increased prod of deoxycorticosterone causes fluid retention suppressing renin/angiotensin system thus causing low-renin hypertension
   - Masculinization and virilization as with 21-α-Hydroxylase Deficiency

Rate-limiting enzyme:
Desmolase
Cholesterol side chain cleavage enzyme
CYP 11A1



Progesterone:

- Secretory phase of uterus and mammary glands
- implantation and maturation of fertilized ovum

Aldosterone:

- Principal mineralcorticoid
- Raises blood pressure and fluid volume
- Increases Na+ uptake and K+ excretion

Testosterone

- Produced in Leydig cells when LH receptor increases c-AMP and PKA
- DHT (more potent than Testosterone) produced by Sertoli cells
- Stimulate spermatogensis
- Promotes development of male and secondary sex characteristics
- Promotes anabolism
Masculinization of fetus

Estrogens:

- Control menstrual cycle
- Promote development of female secondary characteristics

Cushing's Syndrome: Hypercortisolism

- Hyperfuctin of adrenal cortex, usually due to adrenocortical tumor
- High cortisol and low ACTH concetration in plasma (high cortisol in urine)
- Glucocorticoid excess leads to protein loss and characteristic fat distribution

Addison's disease: Primary adrenal cortical insufficiency

- Adrenal cortex atrophy due to disease- Mostly by autoimmune destruction
-Aldosterone and cortisol levels are low, and ACTH is high, which does not leadto hyperplasia due to cortex atrophy

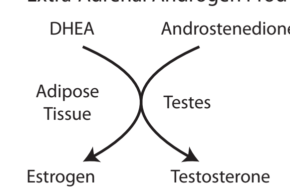
Cortisol:

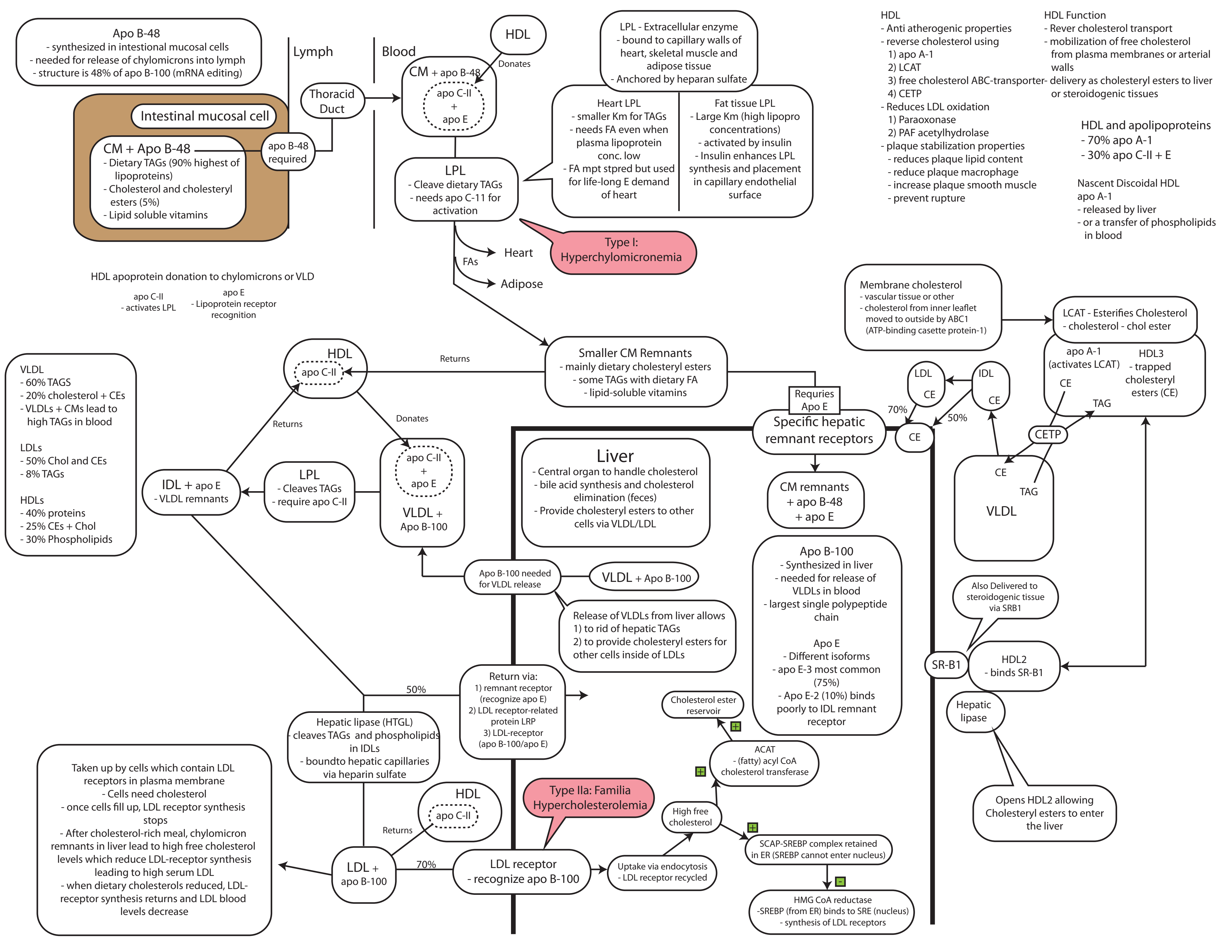
- Dominant glucocorticoid in humans
- Involved in stress
- Needed in adition to glucagon in liver for gluconeogensis
- Protein breakdown in muscle (counteract insulin)
- Anti-inflammatory action
- Bound to transcortin (75%) and albumin (15%)
- Only unbound can enter cells
-areas of inflammation, binding to transcortin is reduced
- Inhibits PLA2 and COX2 - lower eicosanoids

Androgens:

- Produced in Z. reticularis and fasiculata
- Mainly DHEA and androstenedione, but some testosterone

Extra-Adrenal Androgen Prod





Apo B-48
- synthesized in intestional mucosal cells
- needed for release of chylomicrons into lymph
- structure is 48% of apo B-100 (mRNA editing)
Lymph
Blood
HDL
Donates
LPL - Extracellular enzyme
- bound to capillary walls of heart, skeletal muscle and adipose tissue
- Anchored by heparan sulfate
Thoracid Duct
CM + apo B-48
apo C-II + apo E
Intestinal mucosal cell
CM + Apo B-48
- Dietary TAGs (90% highest of lipoproteins)
- Cholesterol and cholesteryl esters (5%)
- Lipid soluble vitamins
apo B-48 required
LPL
- Cleave dietary TAGs
- needs apo C-11 for activation
Heart LPL
- smaller Km for TAGs
- needs FA even when plasma lipoprotein conc. low
- FA mpt stpred but used for life-long E demand of heart
Fat tissue LPL
- Large Km (high lipopro concentrations)
- activated by insulin
- Insulin enhances LPL synthesis and placement in capillary endothelial surface
Type I: Hyperchylomicronemia
FAs
Heart
Adipose
HDL
- Anti atherogenic properties
- reverse cholesterol using
1) apo A-1
2) LCAT
3) free cholesterol ABC-transporter
4) CETP
- Reduces LDL oxidation
1) Paraoxonase
2) PAF acetylhydrolase
- plaque stabilization properties
- reduces plaque lipid content
- reduce plaque macrophage
- increase plaque smooth muscle
- prevent rupture
HDL Function
- Rever cholesterol transport
- mobilization of free cholesterol from plasma membranes or arterial walls
- delivery as cholesteryl esters to liver or steroidogenic tissues
HDL and apolipoproteins
- 70% apo A-1
- 30% apo C-II + E
Nascent Discoidal HDL
apo A-1
- released by liver
- or a transfer of phospholipids in blood
HDL apoprotein donation to chylomicrons or VLD
apo C-II
- activates LPL
apo E
- Lipoprotein receptor recognition
Membrane cholesterol
- vascular tissue or other
- cholesterol from inner leaflet moved to outside by ABC1 (ATP-binding casette protein-1)
LCAT - Esterifies Cholesterol
- cholesterol - chol ester
apo A-1 (activates LCAT)
HDL3
- trapped cholesteryl esters (CE)
CE
TAG
VLDL
- 60% TAGS
- 20% cholesterol + CEs
- VLDLs + CMs lead to high TAGs in blood
LDLs
- 50% Chol and CEs
- 8% TAGs
HDLs
- 40% proteins
- 25% CEs + Chol
- 30% Phospholipids
HDL
apo C-II
Returns
Smaller CM Remnants
- mainly dietary cholesteryl esters
- some TAGs with dietary FA
- lipid-soluble vitamins
Requries Apo E
Specific hepatic remnant receptors
LDL
CE
IDL
CE
70%
50%
CE
CETP
CE
TAG
VLDL
Returns
Donates
IDL + apo E
- VLDL remnants
LPL
- Cleaves TAGs
- require apo C-II
apo C-II + apo E
VLDL + Apo B-100
Liver
- Central organ to handle cholesterol
- bile acid synthesis and cholesterol elimination (feces)
- Provide cholesteryl esters to other cells via VLDL/LDL
CM remnants + apo B-48 + apo E
Apo B-100
- Synthesized in liver
- needed for release of VLDLs in blood
- largest single polypeptide chain
Apo E
- Different isoforms
- apo E-3 most common (75%)
- Apo E-2 (10%) binds poorly to IDL remnant receptor
Apo B-100 needed for VLDL release
VLDL + Apo B-100
Release of VLDLs from liver allows
1) to rid of hepatic TAGs
2) to provide cholesteryl esters for other cells inside of LDLs
Also Delivered to steroidogenic tissue via SRB1
SR-B1
HDL2
- binds SR-B1
Hepatic lipase
50%
Return via:
1) remnant receptor (recognize apo E)
2) LDL receptor-related protein LRP
3) LDL-receptor (apo B-100/apo E)
Cholesterol ester reservoir
Hepatic lipase (HTGL)
cleaves TAGs and phospholipids in IDLs
- boundto hepatic capillaries via heparin sulfate
ACAT
- (fatty) acyl CoA cholesterol transferase
Opens HDL2 allowing Cholesteryl esters to enter the liver
Taken up by cells which contain LDL receptors in plasma membrane
- Cells need cholesterol
- once cells fill up, LDL receptor synthesis stops
- After cholesterol-rich meal, chylomicron remnants in liver lead to high free cholesterol levels which reduce LDL-receptor synthesis leading to high serum LDL
- when dietary cholesterols reduced, LDL-receptor synthesis returns and LDL blood levels decrease
HDL
apo C-II
Returns
Type IIa: Familia Hypercholesterolemia
High free cholesterol
+
+
+
LDL + apo B-100
70%
LDL receptor
- recognize apo B-100
Uptake via endocytosis
- LDL receptor recycled
SCAP-SREBP complex retained in ER (SREBP cannot enter nucleus)
-
HMG CoA reductase
-SREBP (from ER) binds to SRE (nucleus)
- synthesis of LDL receptors

Lipoprotein (a) or Lp(a)
- similar to LDL but with additional apo(a)
- structure of "kringles"
- apo(a) - glycoprotein covalently linked to apo B-100 via disulfide bond and structural analog to plasminogen
- Apo(a) may compete with plasminogen binding to fibrin thus decreased removal (increase of blood clots

Risk factors for Coronary Heart Disease (CHD)
Total cholesterol : HDL ratio >5
- high LDL : HDL ratio
- TAG : HDL ratio > 4

oxLDL
- formed from LDL by oxidation of phospholipids or apoB-100
- (+) Superoxide, nitric oxide, H_2O_2
- (-) Vit E, Ascorbic acid, B-carotine
- attach macrophage via SR-A (back door)
- converts to foam cell
- fibrofatty atheroma and fatty streaks
- oxLDL accumulate in blood b/c not recognized by LDL receptor, thus plaque formation

LDL -A and B (bad)
- A large and less dense
- B (bad) small and dense, penetrate endothelium easily
- B retained by ECM and proteoglycans
B- can be oxidzed to ox-LDL
-B risk factor for CHD

Hyperlipidemias
- High levels of lipoproteins in serum
- Hereditary or acquired
- Frederickson Classification (Type 1-5)

Hypertriacylglycerolemia
- Lipoproteins with high TAGS (chylomicrons and VLDLs)
- related to reduced LPL
- defective apo C-II
- increase release of VLDL

Acquired onset associations:
- Hypertension
- Untreated diabetes mellitus
- Alcohol abuse
- Usage of oral contraceptives
- Hyperuricemia

Hypercholesterolemia
- HIgh lipoproteins with high cholesterol and cholesteryl esters (LDL and lipoprotein remnants)
- Defective LDL receptors
- apo E deficiency

Acquired onset associations
- Hypothyroidism
- Nephrotic syndrome
- Obstructive liver disease
- Treatment with specific medical drugs

Type I: Hyperchylomicronemia
- high chylomicrons in blood after 12-14 hr fast
- genetic deficiency of LPL or apo C-II
- childhood onset
- eruptive xanthomas on trunk butt or extremmities
- creamy layer on top of blood vial
- lipemia retinalis, hepatosplenomegaly, irritability recurent epigastric pain, increased risk of pancreatitis

Hyperlipidemia Type IIb:
- Familial combined hyperlipidemia
- onset puberty, 1:10
- high LDL and VLDL
- very complex (several genes)

Hyperlipidemia Type IV (common 1:100)
- hyperprebetalipoproteinemia
- High serum VLDL
- Caused by LPL deficiency or over-production of VLDL
- High TAG can lead to pancreatitis

Type IIa: Familial Hypercholesterolemia
- High LDL with normal VLDL
- Defective LDL receptor (autosomal dom)
- hetero 1:500 Adult, risk of coron disease
- Homo 1:mil, child onset, child MI and death
- xanthoma over tendon and xanthelasmas

Type III: Dysbetalipoproteinemia (rare)
- high remnants: IDL and chylomicron
- apo E deficiency
- homozygous for apo E-2
- complex but similar blood choles to Type IIa or IIb
- TAG serum level can be elevated
- palmar xanthomas over elbows and knees
- adult onset with accelerated athersclerosis

Hyperlipidemia Type V:
- Mixed hypertriacylglycerolemia
- High serum VLDL and high chylomicrons
- Creamy layer of top on serum

Hypolipidemias
- uncommon

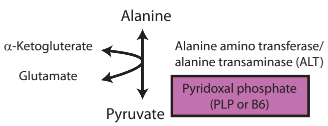
Hypoalphalipoproteinemia
- Low serum HDL
- related to obesity, smoking, some medical drugs and cholesterol reducing drugs

Tangier disease
- hereditary disease
- low serum HDL and coronary heart disease in childhood
- orange tonsils, corneal opacities hepatosplenomegaly
- defective cholesterol ABC transporter
- leads to less substrate for LCAT and early degen of lipid poor apo A-1

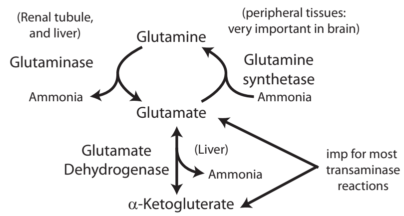
Abetalipoproteinemia
- rare
- low serum VLDL, LDL and chylomicrons
- defect in microsomal TAG transfer protein (MTP)
- MTP needed for formation of VLDL or CM with interaction with apo B
- fat malabsorption
- TAG accumulation in liver and intestine
- retinitis pigmentosa
- peripheral neuropathy

Treatments:
1) Statins
   - Competitive inhibitor drugs
   - Inhibition of HMG CoA reductase
2) Plant stanols and sterols
   - displace cholesterol from micelles during absorption into intestine
3) Ezetimibe
   - reduces cholesterol absorption
   - blocks transporter protein
4) Bile acid sequestering drugs
   - bind bile acids/salts in intestine
   - less dietary lipid digest
   - bile acid/salt excretion in feces
   - low free cholesterol in liver leads more hepatic LDL-receptor synth
   - Thus lower serum LDL

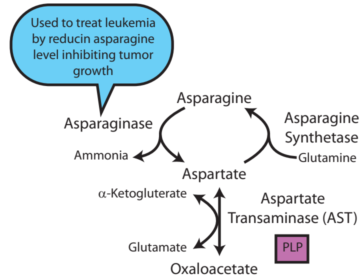
Alanine Metabolism
- Amino group of alanine transferred
- Pyruvate contains keto group
- Transamination is reversible
- Alanine is major transport for amino acid from muscle (especially during starve)
-Alanine (pyruvate) major precursor of gluconeogenesis during starve
- Glucose-alanine cycle involves muscle and liver (where alanine is gen to where gluconeogenesis occurs

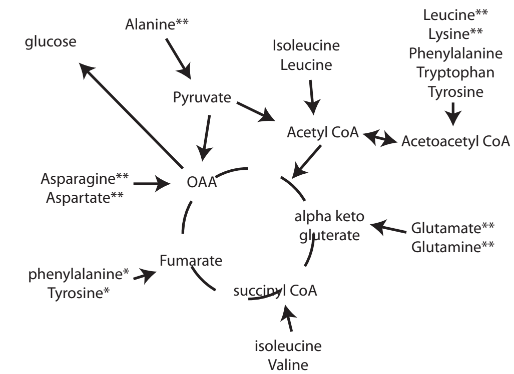


Glutamate Metabolism
- glutamate and glutamine imp for sources of ammonia for urea cycle
- Glut DH does Oxidative deamination (need NAD+) (also reductive amination)
- Glutamine Synthetase imp for scavenging NH3 from circulation (imp in brain and endothelial cells of Hepatic vein)
-Glutamine is non-toxic transport form of NH3
- Renal tissue rich in Glutaminase - acid/base homeostasis



Aspartate metabolism
- Transamination, aspartate to OAA
- Aspartate used for Asparagine synth
- ASN amide deriv of ASP





Nitrogen Metabolism
- needed for:
 1) growth (protein synthesis)
 2) amino acid derivatives
  (Creatine, Purines, heme neurotrasmitters pyrimidines, many others)

Proteosome Deg Pathway (intracellular)

Protein in need of degred
↓
Poly ubiquitinated ATP dependant
↓
Proteosome recognize UB
- protein degraded to peptide
- ATP dependant
↓
UB not recognized thus recycled

Diff Proteins have Dif T1/2

oxidative damage of protein → Aged, damaged protein → destroy old protein synth new

denaturation → Hydrophobic core of proteins gets exposed, (danger of agglutination) → destroy old protein synth new

ubiquitination → UB-chain added to protein, tagging for destruction by proteosome → destroy old protein synth new

PEST sequence → Proteins rich in Pro, Glu, Ser & Thr have shorter t1/2. preferentia UB and destruction → UB-chain added to protein

Tissue Proteins Catabolism/Synthesis (turnover 400g/day) (lysosome and proteosome) ⇄ Amino acid pool (nitrogen pool) - no storage form of nitrogen

Dietary protein (amino acids) → Amino acid pool

Synthesis of non-essential amino acids → Amino acid pool

Amino acid pool → Amino acid catabolism
- Amino acid catabolism → Carbon skeleton → Glucose/lipid synthesis
- Amino acid catabolism → Ammonia (toxic) → Urea cycle → reabsorption → Amino acid pool

Amino acid pool → Synthesis of nitrogen containing compounds: Heme, creatine, neurotransmitters, Purines, Pyrimidines, many others

Tissue protein catabolism
↓
Amino acid
↓ Transamination or deamination — Amino group comes off, or given to another alpha keto acid
α-Keto Acid (Carbon skeleton)
- → Pyruvate/TCA cycle intermediate → Gluconeogensis → Glucose
  - Glucogenic amino acid
  - catabolism yields pyruvate or TCA intermediate (gluconeogensis substrate)
- → Acetyl CoA/ Ketone body
  - Ketogenic amino acid
  - Yields acetoacetate, acetyl CoA or acetoacetyl CoA (not gluconeogensis substrate)

Lysosomal protein degredation
- normal degredation of some cell componants
- Digest from phagocytosisLi
- Digest receptor mediated endocytosis (LDL receptor)
- Autophagy
- Extracellular (sperm to fertilize egg, inappropriate white blood cell lysosome component release and joint tissue damage - BAD)
- lysosomal enzymes stain dark red (work in low pH)

Liver
- Site where most aa release amino groups as ammonia
- → Amino acid carbon skeletons → Gluconeogensis, TCA
- → NH3 (Toxic) → detoxify → Urea (major end product of N metabolism) → Kidney

Kidney
- excrete ammonia as ammonium ions NH4+ (regulation of acid base balance) (Ammonia source -> glutamine, glutaminase enzyme)
-Other non-protein nitrogen secreted:
 1) uric acid (purine deg)
 2) creatinine
 3) other
 (Mostly Urea - 86%)

Cystinuria (COAL - dibasic)
- Tubuluar reabsorption of Cystine decreased (along with ornithin, arginine, lysine)
- inherited deficiency of cystine transporter
- cystine excreted in urine
- cystein precipitates in renal tubules
- renal stones in children

Hartnup's disease
- Defect in transport of neutral amino acid (Tryptophan)
- Decreased tryptophan dietary absorption
- increased tryptophan excretion
- Most patients normal
- May lead to NAD+ deficiency (pellagra) - niacin not synthesized from tryptophan (typically with low protein diet lacking niacin - rare)
4 D's (diarrhea, dermititis, dementia, death)

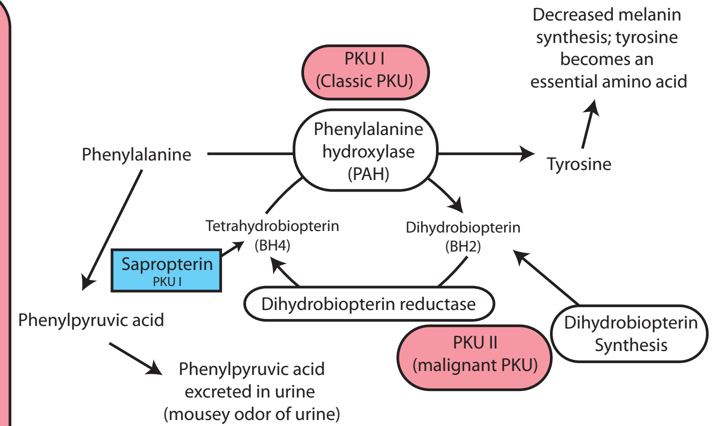
Brush border absprbs AA via secondary active transport (kidney)
-Na+/K+ ATPase (electrochemical gradient)
- Na+ gradient harnessed by some amino acid transport systems

PKU
- Autosomal Recessive and relatively common
- Developmental delay, seizures, spasticity, autistic behaviors hypopigmentation, skin rashes
- Dietary treatment - low protein, no eggs, milk, meat, aspartame
- tested by tandem mass spec (high Phenylalanine levels)
- test infants twice (first may be false negative)

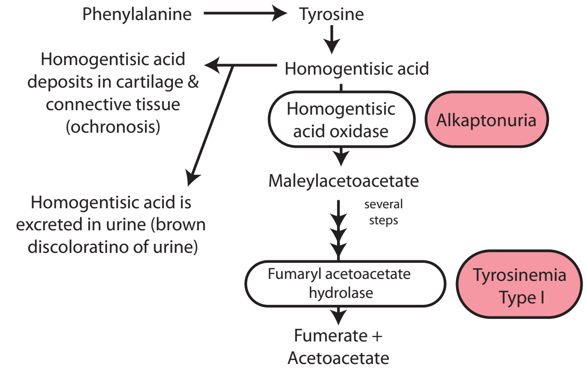
PKU I (Classic PKU)
- CNS: low IQ, seizures if blood Phe high (most severe development seen early in development)
- Phe to phenylacetate and phenyllactate (mousey urine)
- Phe levels elevated
- Decreased Pig. Blonder Blue eyes Pale skin
- Treatment: diet or Sapropterin (synthetic form of BH4 (mild form of disease, may have low affinity mutant enzyme)

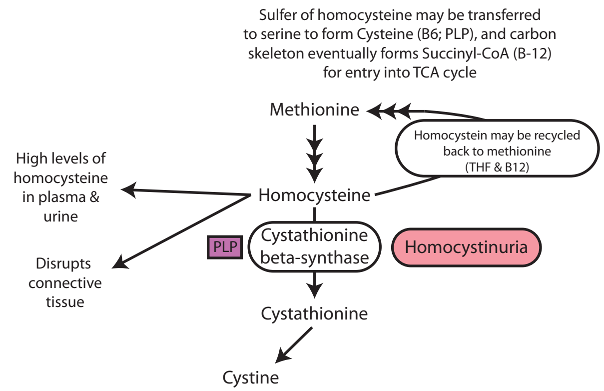
Maternal PKU syndrome
- High maternal blood Phe leads to fetal defects (microcephaly, mental retard, congen heart def)
- High Phe leads to teratogenic properties
- Mom's fault, typically child is heterozygous

PKU II (malgnant PKU)
- deficiant of one of two enzymes for BH2/BH4
- more severe CNS (decreased neurotransmitters: seratonin, dopamine, catecholamines)
- Treatment is diet restriction, biopterin diet and neurotransmitter precursors (difficult because BBB)

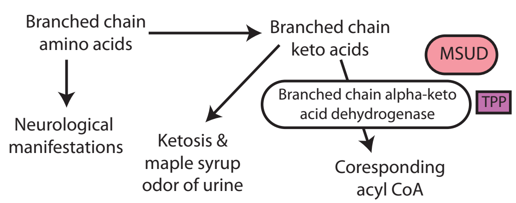


Alkaptonuria
- relatively benign inborn error
- Deficiency in enzyme
- Manifestations:
1) Darkening of urine on standing
2) Discoloration of cartilage and CT (Ochronotic pigments (ochronosis)
3) Leads to severe arthritis (due to oxidation of excess homogentisic acid)
Discoloration: Bluish-black ear, Sclera darkens





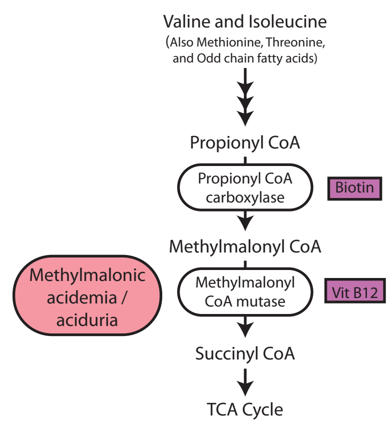
Tyrosinemia Type I
- Buildup of fumaryl acetoacetate
- INborn error of enzyme deficiency
- Manifestations severe and fatal
1) Liver Failure
2) Renal Failure
3) Cabbage like odor of urine
- Dietary restriction of Phe and Tyr may be tried (Difficult because 2 essential aa must be avoided)
- Tyrosine required for NT synthesis (epi, norepi)



Maple Syrup Urine disease (MSUD)
- Relatively rare inborn error of metabolism
- Symptoms develop 4-7 days
-detected by neonatal screening
-Presents with poor feeding, vomiting, poor weight gain and increased lethargy
- Neuro signs develop rapidly:
1) Alternating muscular hypotonia and hypertonia
2) seizures
3) encephalopathy
- Ketosis and characteristic odor present when first symptoms develop
- Coma and death of child in early infancy if not recog and treat
Treatment by restricting branched chain aa.
(difficult and all 3 BCAA essential and 25% aa are VIL)
- Dietary Supplementation with TPP (B1) maybe useful when enzyme has low coenzyme affinity
-Tissue catabolism mobilize aa to metabolism
-Similar to Enzyme similar to Alpha ketoglutarate DH and PDH

Homocystinuria
- Defect in homocystein metabolism thus high in plasmaand urine
- Deficiency of cystathionine beta-synthase (transulfuration pathway)
- Homocysteine binds to CT and disrupts structure
- Characterized by
1) Dislocatino of lens (ectopia lentis)
2) Skeletal abnormalties
3) Mental retardation
4) premature arterial disease
- Some patients respond to oral Vitamin B6

Homocystinuria interferes with collagen and:
1) Lens dislocation after age 3 (other ocular abnormalities
2) Osteoporosis develops during childhood
3) Lipid deposits form atheromas
4) Other effects (lipid oxidation and platelet aggregation -> leading to fibrosis and calcification of atherosclerotic plaques



Methylmalonic aciduria
- Deficient enzyme resulting in elevated levels of methylmalonic acid in circ
- Metabolic acidosis
- Neuro manifestations
1) seizures
2) encephalopathy
- In some children, milder form may manifest
- treatment with B12 (cobalamin) (this form has mutant enzyme with reduced affinity for B12 coenzyme)

Urea Cycle Disorders (UCD)

- Urea formation decreased (hyperammonemia)
- symptoms appear during first few days of life
  1) Lethargy
  2) Irritability
  3) Feeding difficulties
- Results in Neurological manifestations, seizures and mental retardation

Management of Hyperammonemia

1) Dialysis
2) Benzoic acid (conbines with glycine to form Hippuric acid - excreted in urine)
3) Phenylbutyrate - Converted to phenylacetate
   - condenses with glutamine
   - form phenylacetylglutamine
   - excreted in urine
   - Removes 2 N per molec.
3) Low protein high carb diet
4) Prevention of stresses inducing catabolic state
5) Long term: Liver transplant

CPS 1 Deficiency (Type I Hyperammoniemia)**

- Sometimes responds to Arginine intervention (Arg stimulates formation of NAG)
- High levels of NAG might stimulate deficient CPS1

OTC deficiency (X-linked) - (Type II Hyperammoniemia)**

- Most common UCD
- more common and severe in males
- Mitochondrial ornithine transcarbamoylase
- Diagnosed by:
  1) Elevated serum ammonia
  2) Elevated serum & urine orotic acid (orotic aciduria - b/c high carbamoyl phosphate)

ASS - Argininosuccinate synthetase deficiency (Classic Citrullinemia)

- Hyperammonemia with very high levels of serum citrulline, and citrulline in urine
- Treatment may include arginine (Enhances urinary Citrulline excretion) (Activate urea cycle b/c excess substrate)
- Also other listed treatments

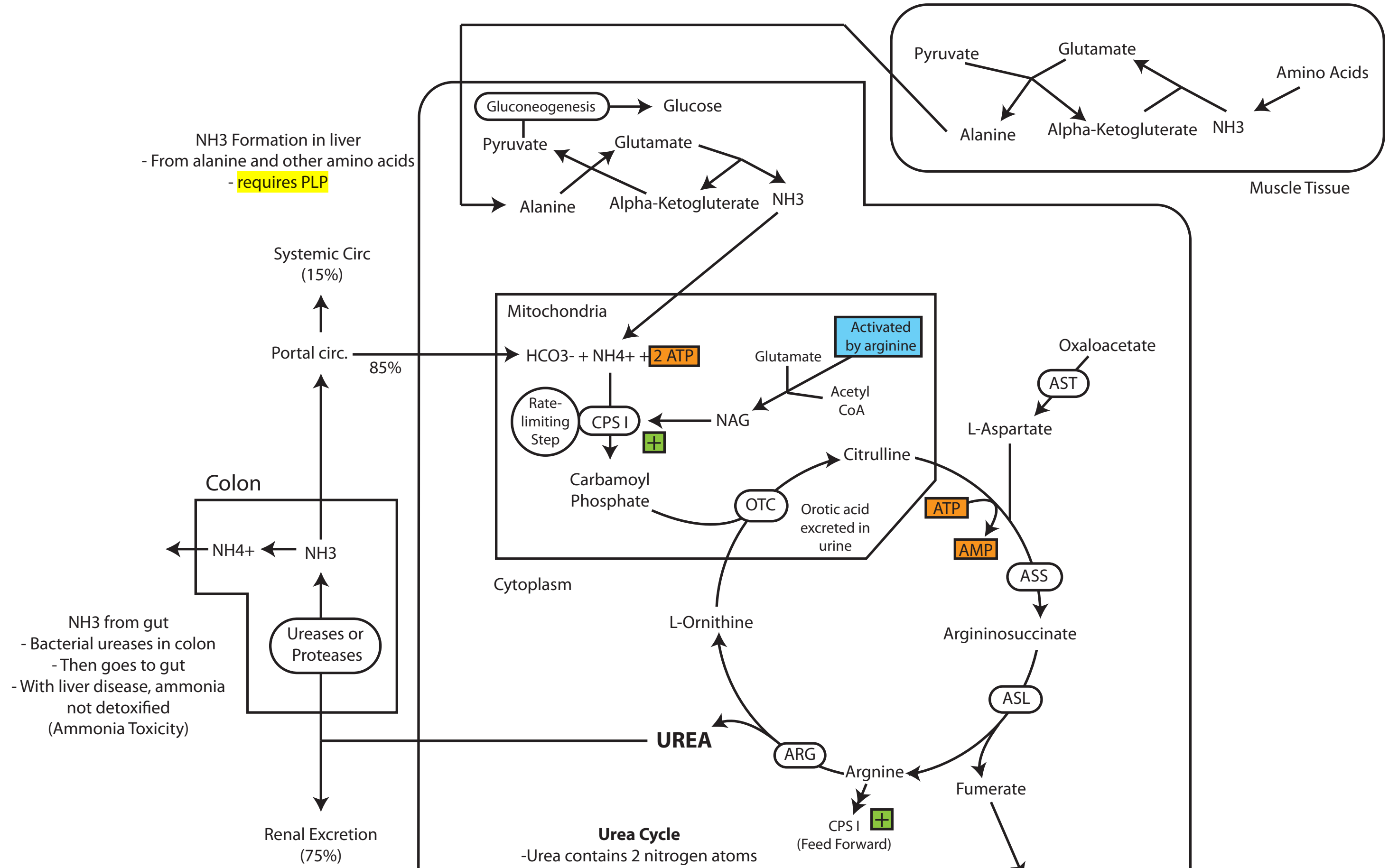
ASL - Argininosuccinate lyase deficiency (Argininosuccinic aciduria)

- Differential diagnosis of hyperammonemia
- Argininosuccinate elevated in plasma & CSF
- moderately high levels of citrulline
- Arginosuccinate detected in urine
- Sometimes treated with surplus of arginine (urinary excretio not argininosuccinate) (May affect Km)
- Treatments listed previously

ARG - Arginase deficiency (Hyperargininemia)

- Diff diagnosis: elevated serum ammonia and arginine (Blood NH3 not as high as other UCD)
- Treat with diet of essential aa excluding arginine
- Other treatments previously listed
- Significantly Adults affected (associated with neurological problems)

** - 1st two enzymes and most severe hyperammonemia



Blood Urea Nitrogen - Kidney Failure
Elevated Blood Ammonia - Urea Cycle Disorder

Acquired Hyperammonemia

- Liver disease due to viral or drug induced hepatitis, alcoholic cirrhosis
- Cirrhosis - porto-systemic shunting of blood
- Ammonia produced in intestine directly enters
- Results in Neurotoxicity

Treatment:
1) Low protein/high carb diet
2) Lactulose (A disaccharide)
   - resistant to digesting in small intestine
   - Bact in colon digest to lactic acid
   - acid neutralized by NH4+
   - More excreted N in feces
3) Neomycin (or other antibiotic treatment) (reduction in bacterial ureases)
4) other previously discussed treatments

Transport of Nitrogen from peripheral tissues

- Ammonia neurotoxic
- Transported as Glutamine or Alanine
- Ammonia not directly excreted in urine (direction excretion resulting in large obligate water loss)

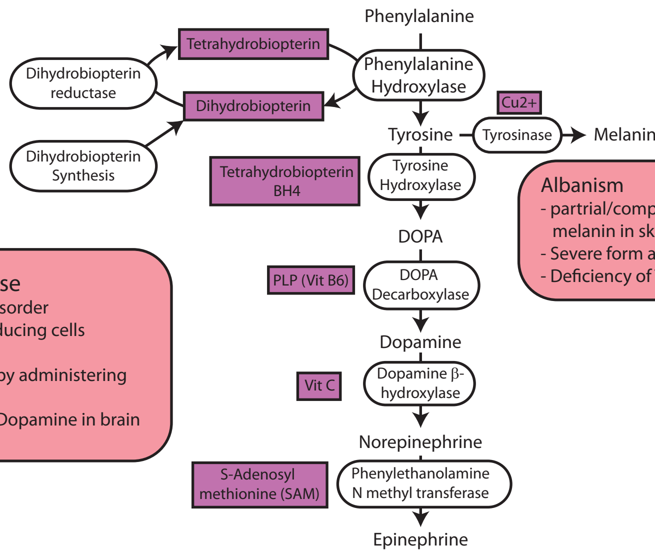
Two hypotheses for mechanism of ammonia neurotoxicity (unknown)

1) Energy and osmotic effect
   - alpha-ketogluterate to glutamate
   - taking TCA intermediates
   - reduced TCA cycle activity
   - reduced ATP (your brain needs ATP
   - Reduced Na+/K+ ATPase activity
   - osmotic pressure off and neuronal cell death
2) Reduced GABA and glutamate levels in brain
   - Glutamate converted to glutamine
   - glutamate needed for formation of GABA
   - GABA is inhibitory NT
   - Glutamine is also major excitatory NT

CPS	Hyperammonemia, Neurological manifestations
OTC (x-linked) (most common)	Hyperammonemia, Increased orotic acid excretion in urine
ASS	Hyperammonemia, Increased citrulline excretion in urine
ASL	Hyperammonemia Increased arginino succinate levels
ARG	Increased arginine levels

Tetrahydrobiopterin deficiency
- Deficiency of dihydrobiopterin synthase or dihydrobiopterin reductase
- hyperphenylalaninemia and decreased neurotransmitter synthesis (catecholamines, serotonin) - results in delayed mental devel and seizures
- Management includes dietary tetrahydrobiopterin

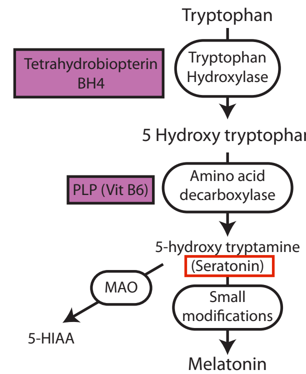
Tetrahydrobiopterin
- Coenzyme required for many AA Hydroxylation reactions
1) Phenyalanine hydroxylase (Phe to Tyr)
2) Tyrosine hydroxylase (Tyr to DOPA)
3) Tryptophan hydroxylase (Trp - 5-hydroxy tryptophan)



Albanism
- partrial/complete deficiency of melanin in skin, hair and eyes
- Severe form affects eyes
- Deficiency of Tyrosinase enzyme

Parkinsons's Disease
- Neurodegenerative disorder
- loss of dopamine producing cells in basal ganglia
- Symptoms improved by administering L-DOPA
- L-DOPA converted to Dopamine in brain

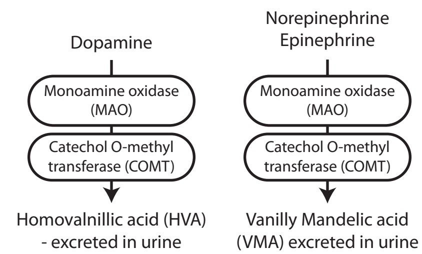
Seratonin Synthesis
- Synthesized in gut, platelets and CNS



Carcinoid syndrome
- Tumor of serotonin producing cells in GIT (APUD cells)
- Cutaneous flushing and sweating
- GI hypermobility -> Diarrhea
- Bronchospasm
- Increased 5-HIAA in urine
- B3 Deficiency

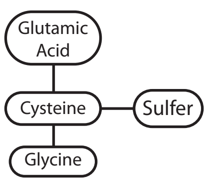
Melatonin Production
- Sleep inducing hormone
- Circadian Rhythm
- Light/Dark cycle
- Seasonal Affective Disorder

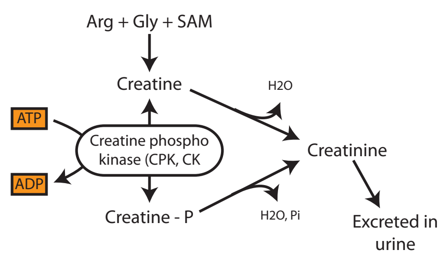
Catecholamine degradation
- Urinary VMA levels must be measured to estimate epi and norepi produced



Pheochyromocytoma: Overproduction of Catecholamines
- High urinary VMA and catecholamines diagnostic
- Adrenal medulla tumor
- Headache, sweating, tachycardia hypertension

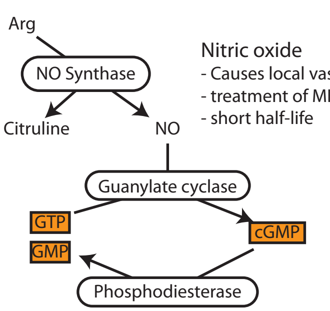
Glutatione
- Intracellular reducing agent (antioxidant)
- Imp for Detox H2O2 especially in RBCs
- Conjugated to drugs to make them more water soluble
- Cofactor for some enzymes
- Aid in rearrangement of protein disulfide bonds





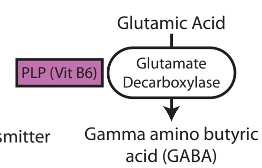
Creatine
- Reservoir of high energy bonds
- Found in muscle, cardiac brain
- Synth from Arg, Glyc & SAM
- Accepts P when muscle resting
- Donates P when muscle contracting
- CPK/CK-MB - MI or ischemia
- Also detect renal failure/kidney function

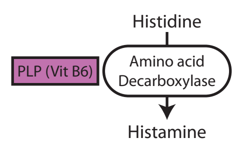
Createinine
- Production is spontaneous
- end product of creatine metabolism



Nitric oxide
- Causes local vasodilation
- treatment of MI
- short half-life

GABA
- inhibitory neurotransmitter in CNS





Histamine
- produced duringn allergic and inflammatory by mast cells
- Vasodilator
- Antihistamine drugs ability of histamine to function as signal
- Receptor antagonist

LIVER

Postprandial (2-3 hrs post meal)
High insulin/glucagon

Substrate availability
- Glycolysis
- Glycogen synthesis
- FA Synthesis
- Cholesterol synthesis

Covalent Modivication by insulin
- Activates protein phosphatases
- dephosphorylation

Enzyme induction by insulin
- Glycolysis (glucokinase, PFK-1, pyruvate kinase)
- PPP (glucose 6-P dehydrogenase)
- FA synthesis (acetyl CoA carboxylase)
- Cholesterol (HMG CoA reductase)

-VLDLs Synth
- TAG Synth
- Glycolytic key Enzyme Synth

Blood Glucose → (GLUT-2) → Glycolysis → PPP, Glycogen synthesis

Glycolysis → PDH → TCA → ETC

TCA → Citrate Lyase forms acetyl CoA in cytosol

→ FA Synth, Acetyl CoA Carboxylase dephosphorylated

→ Cholesterol synth, HMG CoA reductase dephosphylated

Favored at high very high blood glucose levels and following extensive glycolysis

Postabsorptive (5-7 hours)
Low insulin/glucagon
(prevent further drop of blood glucose)

Substrate availability
- B-oxidation
- Ketone Body Synth

Covalent Modivication by Glucagon and Epinephrine
- Protein Kinase A (cAMP)

Enzyme induction by Glucagon
- PEP carboxykinase
- fructose 1,6 bisphosphatase
- glucose 6-phosphatase

FA uptake & B-Oxidation → Acetyl CoA

FA uptake & B-Oxidation → NADH FADH2 → ETC

Acetyl CoA → Pyruvate carboxylase (Needs acetyl CoA) → Gluconeogensis

Glycogenolysis, Gluconeogensis → (GLUT-2) → Glucose Released into blood

Starvation
- glucose into blood by only gluconeogensis
- Glycogen stores decline after 4-6 Hours
- Empty after about 1 day
- Synthesis and release of Ketone Bodies

Hormone Release

Insulin Stimulation (+)
- Glucose
- Amino acids
  - Leucine (dietary essential)
  - Arginine

Glucagon Stimulation (+)
- Epinephrine
- Amino Acids
  - Alanine
  - Arginine

Insulin inhibition (-)
- Epinephrine

Glucagon Inhibition (-)
- Glucose

Stress and Cortisol Release
- Stress activiates ACTH release from pituitary gland
- ACTH stimulates Cortisol from Adrenal Cortex
- Cortisol causes methylation of Norepi and epi (PNMT using SAM) and release from adrenal medulla.
- Epinephrine inhibits insulin release but stimulates glucagon

Muscle Postprandial
- Insulin
- Glucose uptake (GLUT-4)
- Glycogen synthesis
- Glycolysis

- Amino Acid uptake (protein synthesis)

-Usage of Branched-chain amino acids

Exercising Muscle
- Prolonged exercise or starvation
  -> low glucose, so high insulin and high glucagon
  -> FA usage favored
  -> HSL activated
- Contracting muscle
  -> Elevated AMP signals AMPK
  -> Overides other signals
  -> mobilizes GLUT 4 transporters

Skeletal muscle
- Requires ability for anaerobic contraction
  -> >1min for increased Cardiac output
- Lactate produced
  -> Used by resting skeletal and heart
  -> Lactate into Cori Cycle
- Glycogen use during anaerobic muscle cont
  -> 3 ATP rather than 2 ATP
  -> IIB fibers have higher levels of glytolytic enzymes to compensate
  -> Glycogen used 12x faster than Type I
  -> Glyocogen exhausted < 2min anaerobic

Creatine phosphate
- Reservoir of high energy phosphate bonds
- Resting muscle
  -> High ATP levels drive synthesis of creatine phosphate
- Exercising muscle
  -> Low ATP allow creatine-P to drive synthesis of ATP from ADP

Muscle Postabsorptive
- Glucagon, Epinephrine and Cortisol
- Protein degredation
- AA release (alanine-glucose cycle)
- Usage of ketone bodies (Energy)
- Usage of FA (Energy)

Heart muscle
- Continuous active
- Completely aerobic
- negligible glycogen and lipid
- may use any fuel
- Always dependent on vascular' ' supply (interruption = infarc)

Property	Type I slow twitch	Type IIA Fast Oxidative	Type II B Fast glycolytic
ATP Hydrolysis	Slow	Fast	Fast
Contraction Speed	Slow	Fast	Fast
Glycolytic capacity	Low	Moderate	HIgh
Oxidative capacity	High	Moderate	Low
Glycogen storage	+	++	+++
Appearance	Red	Red	White
Capillary supply	Good	Moderate	Poor

Ca2+ and AMP glycolysis and glycogen degredation Regulation
- AMP
  -> stimulates PFK-1
  -> Stimulates glycogen phosphoylase b (only in muscle)
- Ca2+
  -> stimulates glycogen phosphorylase kinase
  -> full activation of glycogen phosphorylase
- Epinephrine
  -> stimulates cAMP cascade in muscle
  -> leads to phosphylation and activation of glycogen phosphorylation
  -> Epinephrine slower than AMP and Ca2+

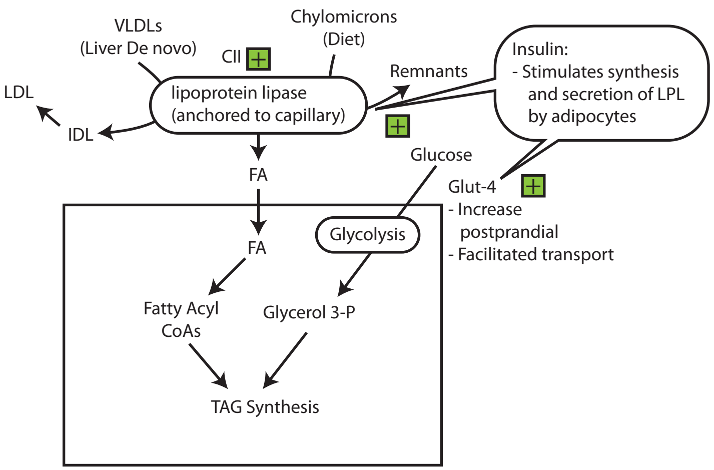
Adipocytes
- 83% total energy
- lipolysis vs. Estereification
  -> determines level of FA in blood
- influences metabolism in other tissue
  -> i.e. liver, skeletal muscle, heart
- Have all enzymes for FA Synth
  -> Only minor amount in humans

Abdominal White Adipose Tissue
- antilypolytic effect of insulin is low
- Very responsive to epinephrine
  -> releases more FA
- Extra FA released goes to liver
  -> leads to increased VLD

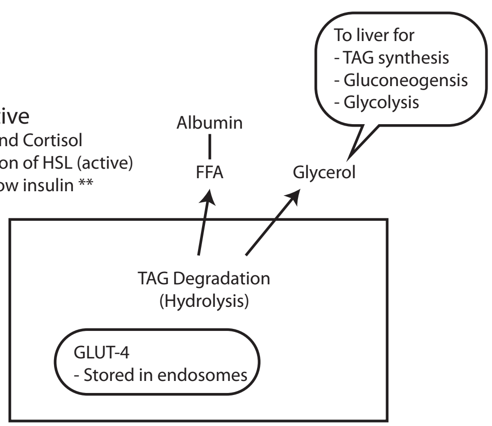
White Adipose
- distributed throughout ody
- Tissue reclaims mobilized FA
  -> Limited amount of FA in blood
- Regulated E storage

Brown Adipose tissue
- Large # mitochondria
- Multiple fat droplets (increased SA and regular cytoplasmic dispersion)
- Thermogenin (UCP-1)
  -> mitochondrial uncoupler
  -> 2,4 DNP - "Death Diet Drug"
- High activity of ETC -> heat production

Posprandial
- Insulin mobilize GLUT-4
- Activate lipoprotein lipase in capillaries
- Lipogenesis (esterification)



Postabsoprtive
- Epinephrine and Cortisol
- Phosphorylation of HSL (active)
- Importantly Low insulin **



TAG mobilization
- Sympathetic nerv sys
- Muscle activity
  -> leads to increase circulation
  -> increased E demand
- Type I Diabetes
  -> Ketone bodies
  -> tissues think body starving
- Elevated cAMP
- Activated PKA

Leptin
- Receptors in hypothalamus
  -> aptetite suppression
- Product of Ob gene
- Satiation
  -> obesity not b/c low leptin

AMPK
- Response to low E states
- many substrates
  -> regulates lots of stuff
- Catabolic and Anabolic

UCP-2
- Sweaty restaurant effect
- Energy dissipation
- Futile cycle

Adipose endocrine gland:
- excess adipose -> insulin resistance
  -> altered adipokine leads to insulin leads to insulin resistance
  -> hyperinsulemia -> pancrease wears down
  -> metabolic syndrome
- Absence of adipose tissue -> severe insulin resistance
  -> Loss of hormonal signals
  -> Dysregulation of TAG and FA levels

Odd Chain FA

Degradation of:
1) Methionine
2) Valine
3) Isoleucine
4) Throenine
go through this same pathways
- Require B12
(AA that form Succinyl CoA produce propionyl-CoA first)

Propionyl CoA
↓ Propionyl CoA carboxylase [Biotin]
D-Methylmalonyl-CoA
↓
L-Methylmalonyl CoA
↓ Methylmalonyl CoA mutase [Vitamin B12]
Succinyl CoA

Cobalamin (B12)

- From meat
- Needed by:
1) Methionine synthase
2) Isomerization of methyl malonyl CoA (mutase)

Deficiency of B12

- Results in accumulation of homocysteine and trapping of THF in methyl-THF form
- Decreases formation of other THF derivatives
- Impairs DNA, RNA and nucleotide synthesis

N10-Formyl THF → Purines
↓ THF
Thymidine
↓ THF
Methionine

Folic acid in Nucleotide and Amino Acid Metabolism

- Requires 2 NADPH

Resynthesis of methionine requires B12 and THF

N5-methyl H4folate or betaine → Methionine Synthase [Vitamin B12] → THF (H4Folate) or N,N-dimethylglycine

Homocysteine → Methionine

Methionine
↓
S-Adenosylmethionine (SAM)
↓
S- Adenosyl homocysteine (SAH)
↓
Homocysteine
↓ Cystathionine Synthase [PLP, B6]
Cysteine

Required for methylation (CH3) reactions:
1) Norepi -> Epi
2) PE -> PC
3) mRNA methylation (methyl-G-cap)

-Produce taurine (bile salt) by cysteine dioxygenase
- Produce PAPS (GAG Synthesis through prod of Sulfate

Vitamin defieicncies associated with Homocystinuria

- Folate + Vitamin B12 - required for homocysteine N-methyltransferase (methionine synthase)
- Vitamin B6 - required for cysthionine synthetase and cysthionine lyase
- Vitamin Deficiency resulting in Increase plasma homocysteine, will damage blood vessels and poses a risk for thrombosis

Clinical correlations of B12:

- Deficiency causes accumulation of fatty acids - > neurological effects (CNS)
- Pernicious anemia - (autoimmune destruction of parietal cells)
- Treatment - lifetime B12 injection
- Can produce megaloblastic anemia
- Anemia is reversible
- CNS effects are not
- Causes of B12 deficiency:
1) Pure vegan diet
2) chronic pancreatitis (can't cleave off R-factor)
3) terminal ileal disease (Chron's disease)

Hyperhomocysteinemia

- deficiency of cystathione synthase
- lens dislocation after age 3 (other ocular abnormalities
- Osteoporosis during childhood
- Mental retardation
- lipid deposits form atheromas
- Lipid oxidation and platelet aggregation -> leads to fibrosis and calcification of atherosclerotic plaques

Folic Acid:

- THF is active form
- Synthesized in bacteria and plants
- essential vitamin in humans
- Important in 1C transfer reactions
1) DNA and RNA synthesis (purines and thymidine)
2) Some amino acid metabolism (glycine <-> serine)
3) Met from homocystine

Serine ↔ Serine hydroxymethyl-transferase → THF
Serine hydroxymethyl-transferase → Glycine, N5,N10-Methylene-tetrahydrofolate

Folate Deficiency Causes:

1) Diet lacking fruit and vegetable
2) Drugs: Methotrexate (chemotherapeutic) trimethoprim (kills blast cells and causes macrocytic anemia)
3) Alcohol (block reabsorption of monoglutamate in jejunum
4) Rapidly-growing cancers (malignant cells use more folic acid)
5) Celiac disease
6) Sulfa drugs

Folic acid Deficiency:

- Most common vitamin deficiency in U.S.
- Common in pregnant women and alcoholics
- reduces risk of neural tube defects
- Results in homocystinuria and megaloblastic anemia

Histidine → Amino Acid Decarboxylase [PLP, B6] → Histamine
Histidine
↓ Histidinase [Histidinemia]
Urocanic acid
↓
N-Formimino-glutamate (FIGlu)
↓ THF
Glutamate

Degredation of Histidine requires THF

- Deficiency in folic acid results in high FIGlu in urine
- Used to diagnose low folic acid
- Histidinemia due to histidinase deficiency

PABA + Pteridine precursor → Dihydropteroate synthetase (Microorganisms) → Folic Acid → Dihydrofolate reductase (Humans and microorganisms) → THF → Amino acid synthesis, Purine Synthesis, Thymidine monophosphate synthesis

Glutamate → (Folic Acid)

Sulfonamides (−)
- Competatively inhibit folic acid in microorganisms

Methotrexate (−)
- Comp inhibit
- Folic acid analogue
- treat cancer

Cancer cells can become resistant to MTX by amplification of gene for dihydrofolate reductase
- MTX and aminopterin affect any rapidly growing cells
- Side effects: anemia, scaly skin, GI tract, respiratory tract disturbances, and baldness

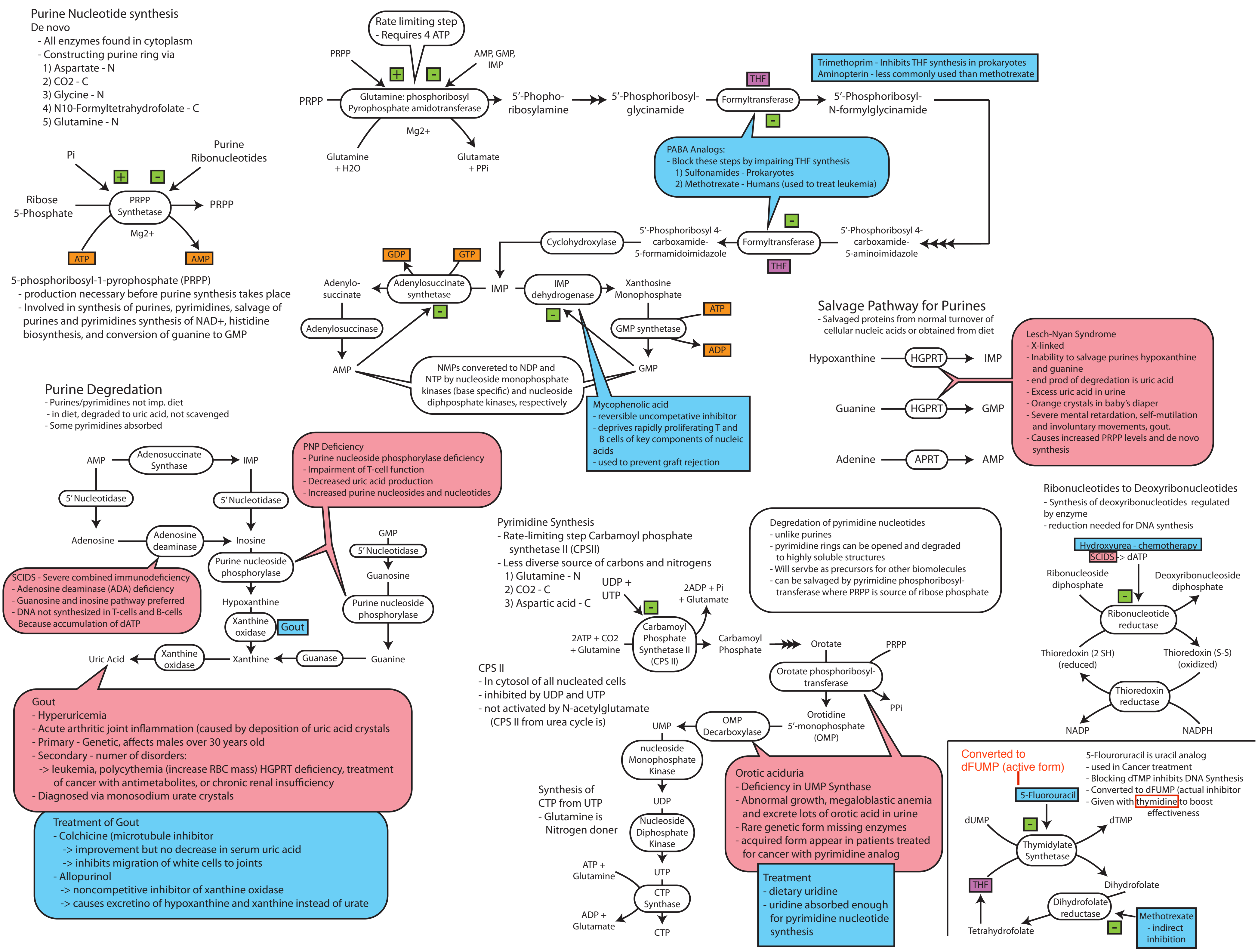
Megaloblastic anemia

- First sign of Folate deficiencies
- Hemoglobin levels low and bone marrow shows abnormally high numbers of megaloblastic cells (large abnormal immature erythrocytes

Folate and B12 deficiency Summary

Folate:
- Megaloblastic anemia
- Homocystinemia, increased risk for cardiovascular disease
- FIGlu accumulation
- Deficiency is important in pregnancy, can occur in alcoholics and malnourished individuals
- Develops in 3-4 months

Vitamin B12
- Megaloblastic anemia
- Homocystinemia, increased risk for cardiovascular disease
- Progressive peripheral neuropathy
- Methymalonyl aciduria
- Pernicious anemia, gastric resection, chronic pancreatitis, vegan, malnutrition
- Develops in years



Purine Nucleotide synthesis
De novo
- All enzymes found in cytoplasm
- Constructing purine ring via
1) Aspartate - N
2) CO2 - C
3) Glycine - N
4) N10-Formyltetrahydrofolate - C
5) Glutamine - N
Rate limiting step
- Requires 4 ATP
PRPP
AMP, GMP, IMP
PRPP
Glutamine: phosphoribosyl Pyrophosphate amidotransferase
Mg2+
Glutamine + H2O
Glutamate + PPi
5'-Phospho-ribosylamine
5'-Phosphoribosyl-glycinamide
THF
Formyltransferase
5'-Phosphoribosyl-N-formylglycinamide
Trimethoprim - Inhibits THF synthesis in prokaryotes
Aminopterin - less commonly used than methotrexate
PABA Analogs:
- Block these steps by impairing THF synthesis
1) Sulfonamides - Prokaryotes
2) Methotrexate - Humans (used to treat leukemia)
Pi
Purine Ribonucleotides
Ribose 5-Phosphate
PRPP Synthetase
PRPP
Mg2+
ATP
AMP
5-phosphoribosyl-1-pyrophosphate (PRPP)
- production necessary before purine synthesis takes place
- Involved in synthesis of purines, pyrimidines, salvage of purines and pyrimidines synthesis of NAD+, histidine biosynthesis, and conversion of guanine to GMP
Cyclohydroxylase
5'-Phosphoribosyl 4-carboxamide-5-formamidoimidazole
Formyltransferase
THF
5'-Phosphoribosyl 4-carboxamide-5-aminoimidazole
GDP
GTP
Adenylo-succinate
Adenylosuccinate synthetase
IMP
IMP dehydrogenase
Xanthosine Monophosphate
ATP
GMP synthetase
ADP
Adenylosuccinase
AMP
GMP
NMPs convereted to NDP and NTP by nucleoside monophosphate kinases (base specific) and nucleoside diphphosphate kinases, respectively
Mycophenolic acid
- reversible uncompetative inhibitor
- deprives rapidly proliferating T and B cells of key components of nucleic acids
- used to prevent graft rejection
Salvage Pathway for Purines
- Salvaged proteins from normal turnover of cellular nucleic acids or obtained from diet
Hypoxanthine
HGPRT
IMP
Guanine
HGPRT
GMP
Adenine
APRT
AMP
Lesch-Nyan Syndrome
- X-linked
- Inability to salvage purines hypoxanthine and guanine
- end prod of degredation is uric acid
- Excess uric acid in urine
- Orange crystals in baby's diaper
- Severe mental retardation, self-mutilation and involuntary movements, gout.
- Causes increased PRPP levels and de novo synthesis
Purine Degradation
- Purines/pyrimidines not imp. diet
- in diet, degraded to uric acid, not scavenged
- Some pyrimidines absorbed
AMP
Adenosuccinate Synthase
IMP
5' Nucleotidase
5' Nucleotidase
Adenosine
Adenosine deaminase
Inosine
Purine nucleoside phosphorylase
PNP Deficiency
- Purine nucleoside phosphorylase deficiency
- Impairment of T-cell function
- Decreased uric acid production
- Increased purine nucleosides and nucleotides
GMP
5' Nucleotidase
Guanosine
Purine nucleoside phosphorylase
SCIDS - Severe combined immunodeficiency
- Adenosine deaminase (ADA) deficiency
- Guanosine and inosine pathway preferred
- DNA not synthesized in T-cells and B-cells Because accumulation of dATP
Hypoxanthine
Xanthine oxidase
Gout
Uric Acid
Xanthine oxidase
Xanthine
Guanase
Guanine
Gout
- Hyperuricemia
- Acute arthritic joint inflammation (caused by deposition of uric acid crystals
- Primary - Genetic, affects males over 30 years old
- Secondary - numer of disorders:
-> leukemia, polycythemia (increase RBC mass) HGPRT deficiency, treatment of cancer with antimetabolites, or chronic renal insufficiency
- Diagnosed via monosodium urate crystals
Treatment of Gout
- Colchicine (microtubule inhibitor
-> improvement but no decrease in serum uric acid
-> inhibits migration of white cells to joints
- Allopurinol
-> noncompetitive inhibitor of xanthine oxidase
-> causes excretino of hypoxanthine and xanthine instead of urate
Pyrimidine Synthesis
- Rate-limiting step Carbamoyl phosphate synthetase II (CPSII)
- Less diverse source of carbons and nitrogens
1) Glutamine - N
2) CO2 - C
3) Aspartic acid - C
UDP + UTP
2ADP + Pi + Glutamate
2ATP + CO2 + Glutamine
Carbamoyl Phosphate Synthetase II (CPS II)
Carbamoyl Phosphate
Orotate
PRPP
Orotate phosphoribosyl-transferase
PPi
Orotidine 5'-monophosphate (OMP)
OMP Decarboxylase
UMP
CPS II
- In cytosol of all nucleated cells
- inhibited by UDP and UTP
- not activated by N-acetylglutamate (CPS II from urea cycle is)
nucleoside Monophosphate Kinase
UDP
Nucleoside Diphosphate Kinase
UTP
Synthesis of CTP from UTP
- Glutamine is Nitrogen doner
ATP + Glutamine
CTP Synthase
ADP + Glutamate
CTP
Degradation of pyrimidine nucleotides
- unlike purines
- pyrimidine rings can be opened and degraded to highly soluble structures
- Will servbe as precursors for other biomolecules
- can be salvaged by pyrimidine phosphoribosyl-transferase where PRPP is source of ribose phosphate
Orotic aciduria
- Deficiency in UMP Synthase
- Abnormal growth, megaloblastic anemia and excrete lots of orotic acid in urine
- Rare genetic form missing enzymes
- acquired form appear in patients treated for cancer with pyrimidine analog
Treatment
- dietary uridine
- uridine absorbed enough for pyrimidine nucleotide synthesis
Ribonucleotides to Deoxyribonucleotides
- Synthesis of deoxyribonucleotides regulated by enzyme
- reduction needed for DNA synthesis
Hydroxyurea - chemotherapy
SCIDS -> dATP
Ribonucleoside diphosphate
Deoxyribonucleoside diphosphate
Ribonucleotide reductase
Thioredoxin (2 SH) (reduced)
Thioredoxin (S-S) (oxidized)
Thioredoxin reductase
NADP
NADPH
Converted to dFUMP (active form)
5-Fluorouracil
5-Flourouracil is uracil analog
- used in Cancer treatment
- Blocking dTMP inhibits DNA Synthesis
- Converted to dFUMP (actual inhibitor
- Given with thymidine to boost effectiveness
dUMP
dTMP
Thymidylate Synthetase
Dihydrofolate
THF
Dihydrofolate reductase
Methotrexate - indirect inhibition
Tetrahydrofolate

Heme Synthesis
- Bone marrow (85%) and Liver
- Heme needed for hemoglobin and myoglobin
- Cytochrome p450 enzymes need heme as prothetic group (liver)
- other enzymes that need heme
  -> cytochromes of ETC
  -> catalase
  -> nitric oxide synthase
- All reactions irreversible

Heme in liver
- tightly regulated (b/c damaging)
- Drugs increase ALAS1 activity as they lead to CYP450 which needs heme
- Synthesis of ALAS1 stimulated at low intracellular heme

Heme in erythroid cells
- Connected to protein synthesis of β-globin chains
- stimulated by erythropoietin
- Erythroipoetin released by kidney at low O2
  -> stimulates RBC and hemoglobin synthesis
- Accumulation is important
- Heme synthesis controlled by intracellular iron availability
  -> Low inracellular Iron downregulates ALAS2
- ALAS2 - X-chromosome
  -> Loss of Funct results in X-linked Sideroblastic anemia

Porphyrias
- Chronic or acute
- Can lead to neurologic dysfunction, mental disturbances or photosensitivity
- Hepatic or erythropoietic porphyrias
- accumulation of intermediates of heme synthesis in blood, tissues and urine
- Treatment includes injection of glucose and hemin

Mitochondria

Succinyl CoA and glycine

Heme –
Low levels of intracellular iron –
→ ALAS1 - most cells / ALAS2 - RBCs — PLP (Vit B6)

Isoniazid
- used to treat tuberculosis
- depletes PLP
- vitamin B6 given to patients

Protoporphyrin IX + ferrous iron → Ferrochelatase (Zinc) (–) → Heme

Lead
- Interacts with zinc cofactors of enzymes
- leads to anemia
- High Blood ALA and protoporphyrin IX in blood and urine

↓

2X δ-aminolevulinic acid (ALA)

↓ AlA dehydratase/ Porphobilinogen synthase (–) (Zinc)

ALA dehydratase porphyria
- Autosomal recessive
- high ALA in blood and urine

4X Porphobilinogen

↓ HMB synthase/ PEG deaminase

HMB synthase deficiency
- Autosomal dominant
- AIP
- High ALA and porphobilinogen in blood and urine
- urine changes dark purple

Hydroxymethylbilane (HMB) → Uroporphyrinogen III (co) synthase → Porphyrin ring III/ uroporphyrinogen III (asymmetric pyrrole ring D) → Uroporphyrin III

Porphyrin ring III/ uroporphyrinogen III → Uroporphyrinogen decarboxylase → Coproporphyrinogen III → Protoporphyrin IX

Porphyria Cutanea Tarda: Uroporphyrinogen decarboxylase deficiency

Congenital Erythropoietic Porphyria
If enzyme is missing, HMB will spontaneously form faulty ring

↓

Uroporphyrinogen I → Uroporphyrin I

↓

Coproporphyrinogen I → Coproporphyrin I (accumulate in skin leading to extremely severe and painful photosensitivity)

Acute Intermittent Porphyria (AIP):
- Deficiency of hydroxymethylbilane synthase
- Autosomal dominant
- ALA activity high in liver b/c heme synthesis defective
- High ALA and porphobilinogen in blood and urine
- Change of urine color to dark purple after 24hrs standing (colorless porphobilinogen spontaneously to porphobilin)
- Ab pain, agitated, tachycardia, respiratory problems confusion weakness
- NOT PHOTOSENSITIVE
- occurs in patients treated with drugs stimulating Cytochrome P450 synthesis, or by infections, ethanol abuse...
- Physicians not administer barbituates (stimulation of CYP450)
- Treatment: Pain meds, IV glucose, saline or hemin

Congenital Erythropoietic Porphyria (CEP):
- Deficiency of RBC uroporphyrinogen III synthase
- autosomal recessive trait
- Uroporphyrin I and coproporphyrin I in tissues, blood and urine (RED PISS AND SKIN)
- EXTREMELY PHOTOSENSITIVE
- blisters, poor wound healing, infections, werewolf
- Child onset
- Hemin infusion would NOT HELP
- Treatment: bone marrow transplant

Uroporphyrinogen III synthase deficiency
- autosomal recessive
- CEP
- Most severe photosensitive
- uropophyrin I and coporphyrin I in tissue and urine

Urophyrinogen decarboxylase deficiency
- Autosomal dominant and acquired
- PCT
- Most common
- high red uroporphyrin III in skin and urine

Porphyria Cutanea Tarda (PCT):
- Most common
- Deficiency of uroporphyrinogen III decarboxylase
- Type I - sporadic, 80%
- Type II - familial, autosomal dominant, 20%
- uroporphyrin III found in tissues, blood and urine (RED)
- Can be caused by
  -> hepatitis
  -> ethanol abuse
- Cutaneous leasons, erosions and bullous lesions in sun exposed regions
- expression influenced by hepatic iron overload or sunlight
- Treatment: avoidance of sun-light, alcohol and iron

Jaundice in newborn
- Newborn infnats have low activity of hepatic UDP-glucuronyl transferase
- Increased destruction of RBCs as well

Kernicterus
- High serum bilirubin levels in newborns
- lipid soluble, unconjugated bilirubin passes BBB
- Hypoalbuminemia/low pH, weaken albumin bond with bilirubin
- Salicylates and sulfonamides compete for albumin binding
- Characterized by lethargy, altered muscle tone, high pitched cry, permanent neuro damage

Phototherapy
- 450nm
- converts bilirubin to more polar water soluble isomers

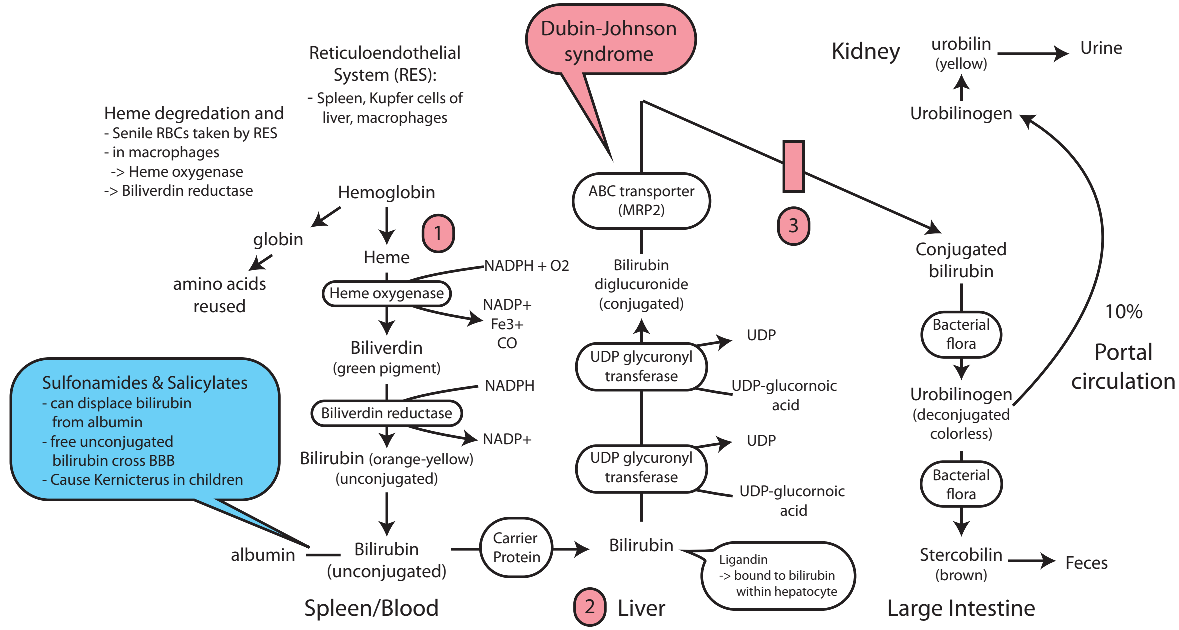
Inherited hyperbilirubinemia
- Crigler-Najjar syndromes I and II, Gilbert's syndrome
- Deficiency of microsomal bilirubin glucuronyl transferase
- unconjugated hyperbilirubinemia

Crigler-Najjar syndrome type I
- Most severe, complete deficiency of enzyme
- high serum bilirubin, kernicterus, mental retardation
- managed by phototherapy, exchange transfusion and prevention of kernicterus
- if not treated, fatal by 2 years

Crigler-Najjar syndrome II (Arias syndrome)
- lower bilirubin glucuronyl transferase activity (10-20%)
- less severe than type I
- Respond to phenobarbital (induces enzyme)
- Phototherapy

Gilbert's syndrome
- mild jaundice following infection, stress or starvation
- UDP-glucuronyl transferase activity 50%
- mild incrase in unconjugated bilirubin

Dubin-Johnson syndrome
- Inherited deficiency of ABC transporter from hepatocyte into biliary canaliculus
- elevated levels of conjugated (direct) bilirubin in circulation



1) Prehapatic Hemolysis

Increased breakdown of RBCs (Hemolysis)
↓
Increased formation of bilirubin
↓
Increased unconjugated bilirubin in circulation (albumin-bound)
↓
Increased uptake and conjugation of bilirubin by liver
↓
Increased excretion of conjugated bilirubin into bile and intestine
↓
Increased excretion of urobilin in urine (yellow)
Increased excretion of stercobilin in feces

2) Hepatic Jaundice "Hepatocellular"

Breakdown of RBCs (hemolysis)
↓
Forrmation of bilirubin
↓
Increased unconjugated bilirubin in circulation (albumin-bound)
↓
Decreased uptake and conjugation of bilirubin by liver & decreased secretion of conjugated bilirubin → Increased conjugated bilirubin in circulation due to regurgitation → Loss of bilirubin in urine (Orange Yellow)
↓
Decreased excretion of conjugated bilirubin into bile and intestine
↓
Decreased formation of urobilinogen in intestine
↓
Decreased excretion of urobilin in urine
Decreased excretion of stercobilin in feces

3) Posthepatic (obstructive) Jaundice

Breakdown of RBCs (hemolysis)
↓
Forrmation of bilirubin
↓
Unconjugated bilirubin in circulation
↓
Normal uptake and conjugation of bilirubin by liver → Increased conjugated bilirubin in circulation due to regurgitation → Loss of bilirubin in urine (Dark Orange Urine)
↓
Obstruction due to gall stones or cancer
↓
Decreased/absent excretion of conjugated bilirubin into bile and intestine
↓
Decreased formation of urobilinogen in intestine
↓
Absent urobilin in urine
Decreased excretion of stercobilin in feces (clay colored feces)

Jaundice
- serum bilirubin >2mg/dL
  -> Normal <1mg/dL (most unconjugated)
- Test via Van den Bergh reaction
  -> Diazo reagent (diazotized sulfanilic acid)
    -- bilirubin reacts with reagent forming red colored complex
- Total Bilirubin = direct (conjugated) + Indirect (unconjugated)
- Unconjugated bilirubin is water insoluble

	Serum Total Bilirubin	Serum conjugated bilirubin	Serum unconjugated bilirubin	Urine bilirubin	Urine urobilinogen
Prehepatic jaundice	Very High	N	Very High	Absent (acholuric jaundice)	High
Hepatic jaundice	Very High	High	High	Present	N or low or high
Post hepatic jaundice	Very High	Very High	N	Present	Low or Absent

α-1 Globulins

Diseases of α-1 antitrypsin
Emphysema
- due to low α-1 AT.
- may be genetic
  -> M allele normal
  -> Z and S allele defective
- Homo - ZZ, high risk developing pulmonary disease and liver disease
- with defect 15% α-1 AT secreted. Rest accumulates in liver (liver disease in children
- Treatment - iv admin of α-1 AT
- Heter normal unless smoking
  -> cigarette oxidizes methionine residues - reduces efficacy

1) α-1 antitrypsin
- Synthesized in liver
- inhibits neutrophil elastase in lungs
- N - glycosylated released into blood and goes to lung

2) α fetoprotein (AFP)
- in fetal plasma
- function similar to albumin in fetal live
- used as marker for cancer of liver ovaries or testes
- Maternal serum AFP
  -> high - fetal neural tube defects
  -> low - down syndrome

3) Transcortin
- Synth in liver
- transport for blood cortisol (75%)
- unbound hormone enters cells reducing inflammation
- reduced binding near inflam sites

4) Retinol binding protein (RBP)
- transports retinol (vitamin A) from liver to peripheral
- Retinyl esters stored in liver

α-2 Globulins

α2 macroglobulin
- protease inhibitor
- one of largest serum protein
- inhibits plasmin and thrombin
- Nephrotic syndrome
  -> 10 fold elevated
  -> large so not lost in urine
  -> albumin smaller so excreted in urine

Ceruloplasmin
- Cu2+ containing plasma protein
- ferroxidase activity
  -> Fe2+ -> Fe 3+ then to transferrin
- Wilson's disease
  -> low levels found
  -> due to less attachment of Cu2+
  -> Kayser-Fleischer Rings
    --> Cu2+ accum in tissue

Haptoglobin
- binds to free hemoglobin (form complex)
  -> from RBC hemolysis
- complex cannot be excreted by kidneys
  -> prevent loss
- Acute hemolysis
  -> Low serum globin
- used to monitor pts with hemolytic anemia

β globulins

Transferrin
- Transports ferric iron between
  -> intest, liver, bone mar., spleen
- can bind 2 Fe3+ atoms
- normally 1/3 binding sites filled with Fe3+
- Patients with iron Deficiency
  -> Low serum iron
  -> Low saturation
- Patients with iron overload
  -> High serum iron
  -> High saturation

Hemopexin
- binds to free heme in blood
- prevents loss of iron by kidneys

β-lipoproteins (LDLs)
- + charge (have only apo B-100)

γ globulins

IgM:
- in blood and lymph
- first antibody responder

IgG:
- in all body fluids
- smallest but most common
- produced on repeated exposure to same antigen (often bacterial and viral infec.
- crosses placenta and confers immunity to fetus and newborn

IgE:
- in lung, skin, mucous membranes
- secreted in allergic reactions

IgA:
- Found in body secretions
- protects body surfaces

IgD:
- role uncertain

Albumin

- smalles most abundant serum proteins
- anionic at pH 7.4 (negative charge)
- 70-80% osmotic pressure
- synthesized by liver @ 14g/day
- t1/2 20 days
- glycosylated in blood - diabetes test?)
- single polypetide chain with 585 aa and stabilizerd by internal disulfide bonds
- indicator of nutritional statis and liver synthetic function
- low albumin indicates severe liver damage/inflammation
- Albumin in urine
  -> damage to kidney
- binds cations in blood (Ca2+)

PH AND BOUND CA2+
- High pH = High bound Ca2+ Albumin
- Low pH = Low bound Ca2+ Albumin

Kernicterus

- infants with displaced bilirubin
- because albumin tranportsbilirubin and drugs
- thus aspirin will displace bilirubin for drugs

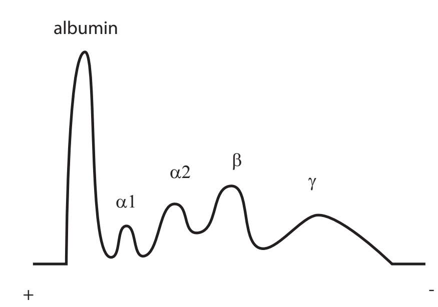
Edema
- low serum albumin

hypoalbuminemia
- low colloid osmotic pressure
- Congenital analbuminemia appear normal and only show some edema or hypocalcemia

Different causes of hypoalbuminemia
- Decreased albumin synthesis
  -> low protein diet (malnutrition)
  -> Chronic liver disease (cirhosis)
- Increased albumin loss
  -> severe burns (loss from burnt skin)
  -> Loss of albumin in urine (nephrotic syndrome)

Serum proteins

- separated by electrophoresis
- Separation by charge (albumin -)



Multiple myeloma

- tumor of plasma cells (activated B cells)
- increase of monoclonal band
- specic immunoglobulin produced by malignant clone
- typically high amounts of monoclonal Ig
- important in diagnosing and monitoring patients with multiple myeloma

Liver Cirrhosis
- increase in many immunoglobulins
- of all closses, often includes IgA

Acute phase proteins

- serum proteins increased (Positive) or decreased (Negative) in response to inflammatory disorders
- found in response to infection, extreme stress, burns, major crush injury allergy or other

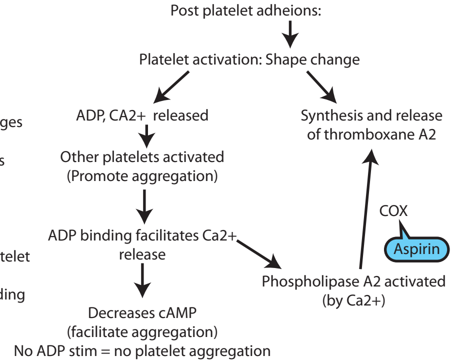
Reactants
- Cytokines
  -> stimulate synthesis of positive acute phase reactants
  -> serum proteins serve immun functions
- Ceruloplasmin and haptoglobin
  -> inhibit iron uptake by microbes
- Leads to increase α2 -globin fraction
- Increase of α-1 AT leads to increase in α-1 Globulin fraction
- NOT in serum
  -> C-reactive protein not found, but released over 30,000 times
  -> need specific test and is used to monitor inflammation

Step 1: Vascular spasm/ vasoconsytriction
- Trauma to vessel wall results in contraction
- factors released from injured vessel wall
- Nervous reflexes
- Transient and not long term cessation of bleeding

2) Platelet plug formation: Primary hemostasis
- Greatly limits blood loss by forming plugs
- 2 steps: platelet adhesion followed by aggregation

Role of Von Willebrand factor
- vWf - bridge between specific GPs
- FAciliatates platelet adhesion and platelet aggregation
- complexes with VIII, carries it, stabilizes it and prevents its degradation
- Deficiency associated with defect in formation of platelet plug (primary hemostasis) and defect in coagulation (due to low VIII)

A) Platelet adhesion
- Facilitated by endothelial injury (normally inhibted by negative charges of wall and platelets)
- mediated by receptors glycoproteins (GPIb and Gpla)
- Steps:
  1) Platelet GPIa binds to collagen and structural changes
  2) Von Willebrand factor binds to platelet receptor GPIb
  3) binding exposes GPIIb/IIIa for binding of fibrinogen



B) Platelet aggregation
- mediated by Fibrinogen
  -> binds to GPIIb/IIIa on adjacent plates
- Thrombasthenia of Glanzman and Naegeli
  -> GpIIb/IIIa defect
- Formation of platelet plug/primary hemostasis

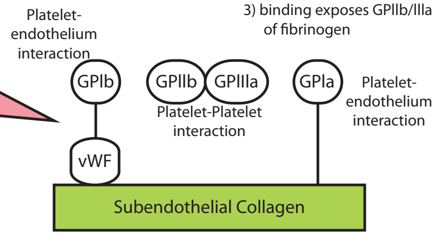
Primary hemostasis - formatin of platelet plug
- Formation of platelet plug requries
  1) von Willebrand factor
  2) adequate number of platelets
  3) GpIb platelet receptors
  4) GpIIb/IIIa platelet receptors
- Defects in platelet plug formation
  -> because of missing any of above
  -> increased bleeding time

DRUGS!!
- Aspirin and COX1 inhibitors
  -> prevent formatino of thromboxane in platelets
- Heparin
  -> activates antithrombin III
  -> inactivates thrombin
- Warfarin
  -> blocks epoxide redutase in liver
  -> thus preventing regen of active form of Vit K
  -> also inhibits clottin factors (against action of Vit K
- Streptokinase
  -> thrombolytic agent; plasminogen activator
  -> converts plasminogen to plasmin enabiling dissoltion of clots

Platelet defects: Increased Bleeding time (due to defective platelet plug formation)
- GpIb defect: Bernard-Soulier syndrome
- Normal platelet count and increased bleeding time
- Qualitative platelet defects

GpIIb/IIIa defect: Thrombasthenia of Glanzman and Naegeli
- Normal platelet count and increased BT
- Also qualitative platelet defect

Thrombocytopoenia (low platelet count
- low platelet count and increased BT
- Quantitative platelet defect



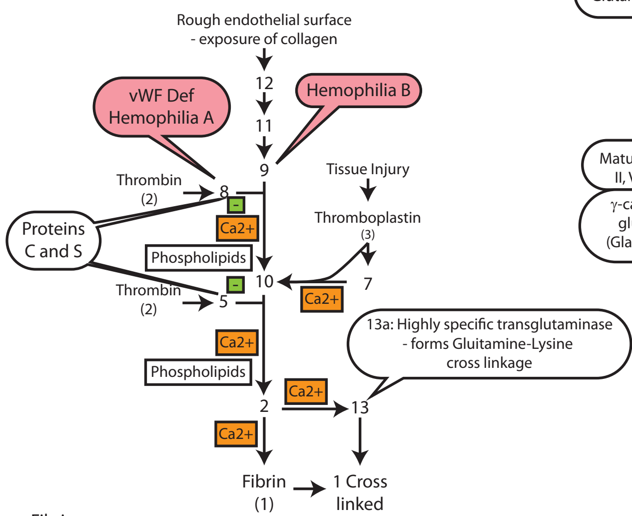
Control of hemostasis:
- Normal endothelia are anti-thrombic
- chemical mediators
  ->PGI2 and nitric oxide
  -> released by healthy endothelium
  -> PGI2 increases cAMP levels, inhibing platelet activation

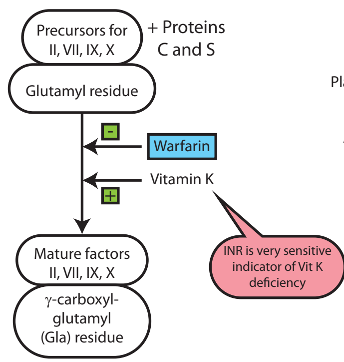
Anti-coagulant factors
- Antithrombin III
  -> inhibits factors Xa and thrombin (IIa)
  -> Heparin works by activating this factor
- Protein C and S (also require vit K)
  -> inactivate cofactors Va and VIIIa
  -> Protein S is cofactor for protein C
  -> Protein C activated by binding thrombomodulin to thrombin

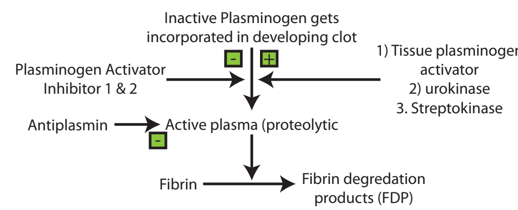
3) Blood coagulation (Secondary hemostasis)
- Conversion of blood from liquid state to solid gel state
- Ultimate goal: soluble fibrinogen to insoluble fibrin and stabilize (via cross linking)
  -> requires thrombin
- 2 pathways
  1) Extrinsic Pathway
    - activated on injury
  2) Intrinsic pathway
    - important pathological processes e.g. atherosclerosis
- Both systems involve cascade
- Product formed at each step catalyzes next step

Role of vitamin K & γ-Carboxylation
- Vit K required for hepatic synthesis of Prothrombin (II), VII, IX, X, Proteins C and S
- glutamic acid residues of above proteins are carboxylated in reaction involving vitamin K
- allows Ca2+ binding b/c of negative charge
  -> forms complex with Ca2+ allowing it to bind to phospholipids on platelet membrane

Step 4: Dissolution of the fibrin clot/ tertiary hemostasis
- Fibrin degredation products and D-dimer levels raised with deep vein thrombosis
- estimated to clinical practice to estimate extent of patients with thrombosis







Tests of hemostatic functions

Bleeding time
- time from initial injury to platelet plug formation
- indicator of platlet plug formation
- Prolonged bleeding time indicator of
  1) Low platelet count
  2) vWf deficiency
  3) Platelet receptor defects

Clotting time
- time formatin of stable fibrin
- Prolonged clotting time indicates defect in coagulation pathway

Prothrombin time (PT) or International normilized ratio (INR)
- Tests extrinsic and common coagulation pathway

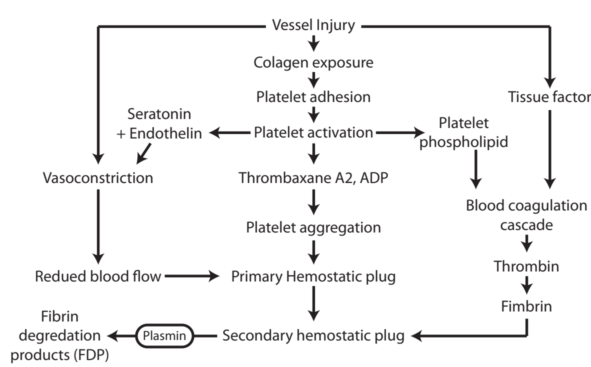
Partial thromboplastin time (APTT)
- Tests intrinsic and common pathways

Fibrinogen:
- Plasma protein synthesized in liver
- Thrombin cleaves fibrinopeptides of fibrinogen forming fibrin
- Fibrin Monomers aggregate, linked via H-bonds (Soft Clot)

Platelet phosphlipids in clotting
- Activation of platelets -> exposure of phospholipids on platelet surface
- Provide binding sites for coagulation factors

"Hard" clot: Via cross linking
- Mediated by Factor XIII activatedby thrombin
- Covalent cross linking bond
- (amino group of lysine from 1 molecule cov link to amide nitrogen of glutamine)
- (XIIIa highly specific transglutaminase)

Overview:



Hemophilia
- inherited coagulation disorder (x-linked)
- Hemophilia A - VIII and Hemophilia B- IX

Bleeding time	Normal (platelet plug formation normal)
Platelet count	Normal
Clotting time	increased (defect in intrinsic pathway)
Prothrombin time (INR)	Normal (extrinsic and common coagulation pathway normal)
APTT (Intrinsic pathway)	increased (defect in intrinsic pathway)

Von Willebrand Disease
- Most common inherited bleeding disorder
- defect in platelet plug formation
- instability of Factor VIII
Presents similar to hemophilia A

Bleeding time	Prolonged (defect in platelet plug formation)
Platelet count	normal
APTT (Intrinsic pathway)	Prolonged (Factor VIII levels may be low normal)
PT (INR)	Normal (extrinsic pathway normal
vWF levels	LOW

Functions of liver:
- excretion, plasma protein synthesis, blood clotting, metabolism, detox
- Used in clinic for liver function tests (LFTs)

Liver function tests (LFTs)
- Serum bilirubin (total, uncon, con)
- Enzymne levels in serum indicating
  -> Hepatocellular damage
  -> Cholestasis
  -> GGT (Drugs or Cholestasis)
- Synthetic function
  -> Albumin
  -> Clotting factors (PT)
  -> alpha feto protein (AFP)
- Tests for metabolic functions
  -> serum ammonia
  -> blood glucose

When do we use LFTs?
- Assess intensity and type of
  -> Jaundice (hepatic or obstructive)
- Assess extent of hepatocyte damage
  -> viruses
  -> malignancy
  -> drugs
- Differentiate between
  -> chronic
  -> acute
- Monitoring drug therapy of hepatotoxic drugs
- Follow up and prognosis of liver diseases

Signs and symptoms during physical exam and possible causes
- Jaundice
  -> acute hepatitis
  -> biliary obstruction
  -> advanced chronic liver disease
- Abdominal pain and fever
  -> cholesystitis
  -> inflammatory liver diseaseo
- Stigmata of chronic cirrhosis
  -> spider angioma
  -> palmar erythema
  -> gynecomastia
  -> testicular atrophy
- complications of liver disease
  -> advanced liver disease
  -> encephalopathy, ascites, GI bleeding (esophageal varicies), coagulation disorders

Serum bilirubin
- estimate extent of liver damage
- difficult to distinguish hepatocellular from cholestatic disorder
  -> elevated in intra and extra hepatic cholestasis
- Intra-hepatic: observed in viral hepatitis, drug toxicity and cirrhosis
- Extra-hepatic: bile duct/gall bladder obstruction or tumors around bile duct/ head of pancreas

Hepatitis
  -> Increased unconjugated (decreased liver uptake
  -> Increased conjugated (associated intra-hepatic cholistasis

Cholestatic disease
- Serum bilirubin mostly elevated
- Complete obstruction
  -> bilirubin regurgitated in blood (in urine)
  -> urobilinogen absent in urine
  -> stercobilin absent in feces (extent of absence depends on severeity)

Serum Enzymes
- Indicator of
  -> Hepatocellular damage
  -> Cholestasis
- degree of enzyme rise corresponds to extent of damage

Enzymes Indicating Hepatocellular Damage:
Aminotransferases (Transaminases)
- ALT (Alanine Aminotransferase)
  -> More specific for liver cell damage
  -> Acute Hepatitis, levels are higher
- AST (aspartate aminotransferase.
  -> Long standing alcoholic cirrhosis
  -> AST:ALT ratio 2:1
- Levels raised in viral hepatitis, drug induced hepatitis, and long standing obstructive jaundice (levels lower than hepatitis)

Enzymes Indicating Cholestasis:
(Intra and extra hepatic)
- ALP (Alkaline phosphatase)
  -> secreted by biliary canaliculi
  -> Elevated with both intra and extra hepatic obstruction
  -> Also Elevated during pregnancy growing children and bone disease
- GGT (Gamma glutamyl transferase)
  -> secreted by biliary ducts
  -> synthesis induced by alcohol and drugs
  -> Used to differentiate between hepatic and non hepatic ALP elevation
- Elevation in both ALP and GGT
  -> Cholestasis usually present
  -> Ultrasound performed to assess Intra or Extra hepatic biliary stasis
  -> Dilated biliary tree = extrahepatic
  -> no dilation = intrahepatic

Test for Liver Synthetic Function
- All plasma proteins except γ-globulins synthesized by liver

Serum Albumin
- Only synthesized in liver

In chronic liver disease
- low albumin is characteristic
- contributes to ascites and pedal edema

Acute hepatitis
- albumin levels normal

Cirrhosis
- γ-globulins usually elevated in

Clotting factors (PT)
- Liver synthesizes clotting factors V, VII, IX, X, protrhombin (II) and fibrin (I)
- Normal prothrombin time: 12-15 s
- Liver also involved in post translational vitamin K dependent γ-carboxylation of glutamic acid
  -> factors II, VII, IX and X
- Important monitor during surgery
- Increase indicates
  -> Hepatocellular disorders (decrease in synthesis
  -> cholestasis (decrease Vit K absorption no bile and vit K fat soluble

Tests evaluating metabolic function

Blood glucose
- liver is major glucostat
- regulation by storing and releasing glucose

Chronic liver disease
- alter blood glucose level

Blood amonia levels
- ammonia detoxified to urea in liver

Fulminant and end stage liver disease
- impairment of urea formation
- blood ammonia levels rise
- High blood ammonia contributes to
  -> Hepatic encephalopathy
  -> altered consciousness

Lipid metabolism
- liver secretes VLDL transporting TAG

Fatty liver
- excessive deposition of TAG in liver
- damage to hepatocytes by ethanol results in fatty liver
- caused by:
  -> high NADH thus low b-oxidation
  -> high acetyl CoA thus FA synthesis
  -> decreased VLDL secretion because hepatotoxic EtOH effect

Special tests evaluating liver function:
- alpha feto protein (AFP)
  -> tumor marker for liver cancer
- Fe, transferrin and ferritin
  -> hemochromatosis
- Ceruloplasmin
  -> Wilson's disease
- α1-antitrypsin
  -> α1-antitrypsin deficiency

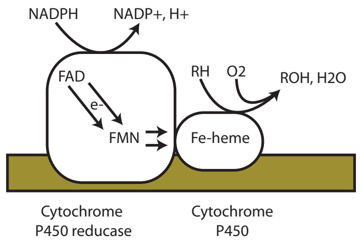
Laboratory findings	Acute hepatocellular damage	Cholestatic jaundice	Chronic alcoholic cirrhosis
Serum total bilirubin	↑↑	↑↑	↑
Serum conjugated bilirubin	↑	↑↑	↑
Serum unconjugated bilirubin	↑	N	↑
ALT	↑↑↑	N/↑	↑
AST	↑↑	N/↑	↑↑
ALP	↑	↑↑↑	↑
GGT	↑	↑↑↑	↑
Serum albumin	N	N	↓

Xenobiotics
- Not normally found in human body
- Drugs, food additives or pollutants
- Primarily modified in liver or in GIT, lungs, brain, kidney, skin
- Drug metabolism
  -> expose functional group (hydroxyl)
  -> makes molecule more polar for excretion in kidney
- Two phases of metabolism

  -> Phase II: usually follows I, but may start here

Phase I: involving cytochrome P450 (CYP 450)
- ER membranes
  -> Hydroxylation or many other reactions by microsomal Cytochrome P450

Phase II:
- Often after Phase I
- In cytosol
  -> UDP-glucuronic acid
  -> PAPS (sulfate)
  -> Glutathione
  -> Amino acids



CYP450:
- Heme containing enzyme induced by EtOH and drugs
- CYP450 reductase works with CYP450
  -> Providing NADPH
  -> two e- taken up and bound to FAD then FMN
  -> Donates 1 e- at a time to heme group of CYP450
- Reaction very complex and radicals generated

Induction of CYP450
- Done by drugs of specific CYP450 that metabolize them
- induction complex, but increase of transcription
- Phenobarbital leads to 3-4 fold increase in 4-5 days
  -> stimulates heme synthesis
  -> drug should be avoided with AIP
  -> bad to elevated ALA levels (similar to GABA)

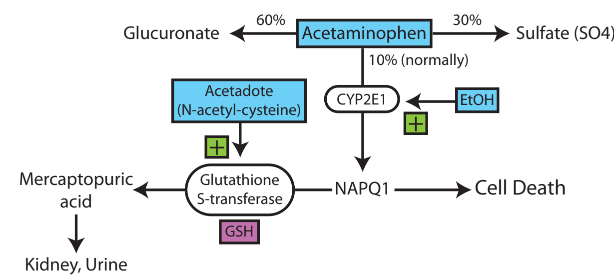
CYP2E1
- Part of MEOS system in liver (microsomal EtOH oxidizing system
- Induced during chronic ethanol consumption

Drug interactions
- Grapefruit juice
  -> inactiavtes CYP3A4
  -> statins less inactiavted by enzyme
  -> leads to liver damage
- Tamoxifen (breast cancer treatment)
  -> needs activation by cytochromes in liver
  -> inhibition of CYP450 leads to less effectiveness

Treatment with more than 1 drug
- One drug can induce CYP3A4 leading to different metabolism of another drug
  -> DANGEROUS!
- Ex. Warfarin less effective with phenobarbital
  -> thus higher dosage of Warfarin for patients perscribed with both
  -> if pt comes off phenobarbital, can lead to excessive bleeding

Different CYP450
- Hundreds of isoforms grouped into 6 major groups
- CYP3A4
  -> 1/3 of CYP450 in liver
  -> acts on more than half of therapeutic drugs
- CYP2E1
  -> specific for EtOH metabolism

Acetaminophen Detoxification
- Normally conjugates with glucuronate or sulfate and excreted in urine
- Occurs to lesser extent, with CYPE1
- With chronic alcoholics
  -> more processed by CYP2E
  -> produces NAPQ1 (TOXIC!) but can still be detox by GSH
- Poisoning treated with acetadote
  -> binds directly to NAPQ1 and provides cysteine for glutathione synth

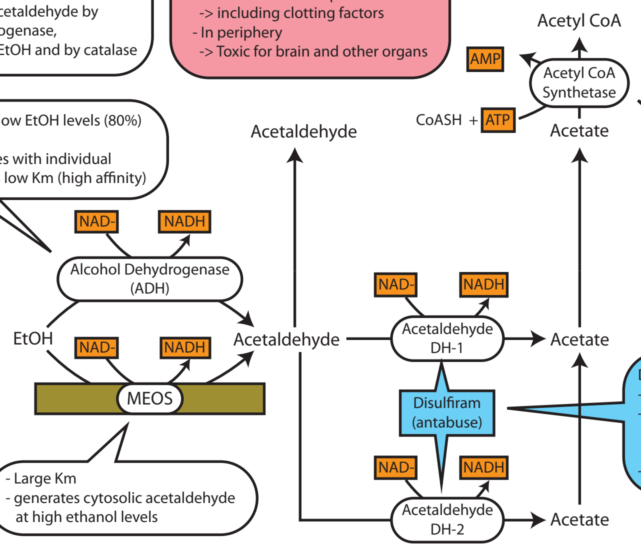


Ethanol Metabolism:
- Uptake starts in stomach
- Efficient on empty stomach (70%)
- Metabolism mainly with hepatocytes
- Metabolized to acetaldehyde, then to acetate
- Metabolized to acetaldehyde by
  ->alcohol dehydrogenase, MEOS at high EtOH and by catalase

Toxicity of acetaldehyde in chronic ethanol abuse in liver:
- Inhibits VLDL release
- Binds to GSH
- Very high Acetaldehyde
  -> inhibits relase of proteins
  -> including clotting factors
- In periphery
  -> Toxic for brain and other organs

- Main enzyme at low EtOH levels (80%)
- Seven isozymes
- Metabolism varies with individual
- liver isozyme has low Km (high affinity)

- Used mainly in muscle
- irreversible
- enters TCA cycle
- Enzyme has low activity in hepatocytes



Disulfam (antabuse)
- inhibits acetaldehyde DH
- Side effects
  -> flushing, nausea
  -> meant to prevent EtOH consump
- Dangerous if large EtOH intake

- Large Km
- generates cytosolic acetaldehyde at high ethanol levels

Mitochondrial acetaldehyde (DH-2)
- small Km, fast reaction
- forms NADH and acetate in mitochondria
- 40% asians have less active form

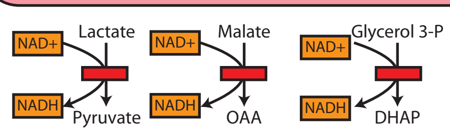
Ctysolic acetaldehyde (DH-1)
- cytosolic isoform
- higher Km, formed in cytosol

Methanol or Ethylene glycol
- Both substrates in cytosol for alcohol DH
- Treatment is EtOH or Fomepizole
  -> comp inhibitor
  -> both have higher affinity
- Also sodium bicarb to reverse acidosis
- Severe cases: hemodialysis needed

Methanol
- oxidzed to form formaldehyde
  -> by Alcohol DH
  -> highly toxic
- prod leads to
  -> mental and visual disturbance
  -> irreversible blindness (severe)

Ethylene glycol
- metabolized to glycoaldehyde
- then to TOXIC oxalate
- antifreeze
- leads to
  -> kidney failure and death
  -> due to formation of Ca oxalate

High cytosolic NADH/NAD+ ratio, gluconeogensis cannot use precursors
- Lactate stays in blood
- Glucogenic aa cannot be used (Malate - OAA)
- Glycerol released by fat, cannot be used for gluconeogensis



only insoluble cofactor

Vitamins:
- chemically unrelated organic compounds
- cannot be synthesized
- supplied by diet
- Perform specific cellular functions
- 13 necessary for health
- classified according to solubility

Fat soluble Vitamins:
- absorption dependent on normal fat digest
- Maldigestion/malabsorption of fat results in secondary deficiency

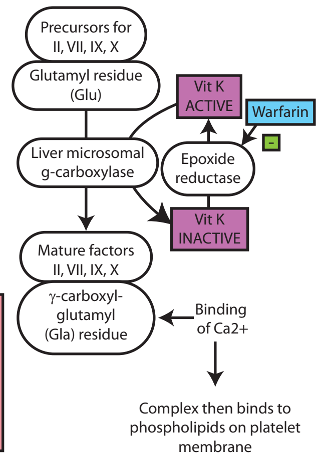
Water Soluble vitamins
- B complex
- C

Vitamin deficiency
- Primary
  -> Dietary deficiency
  -> Starvation
  -> Malnutrition
- Secondary
  -> Reduced intake (dental problems, chronic disease, morning sickness)
  -> Malabsoprtion (diarrhea, genetic defect)
  -> Increased requirements (pregnancy, post op, periods of rapid growth)
  -> increased loss (lactation)

Vitamin K
- Coenzyme for post translational modification of clotting factors
- Diet and synthesized by intestinal bacterial flora
- Forms
  -> Phylloquinone (dietary plant source)
  -> Menaquinone (intestinal bacteria)

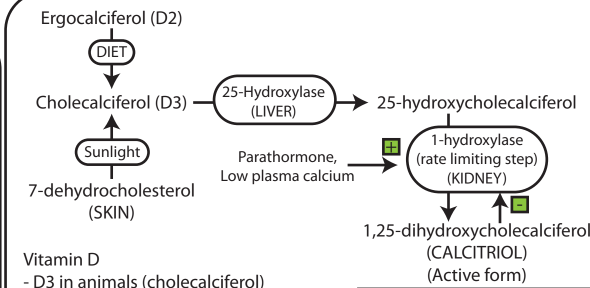
Role of vitamin K & γ-Carboxylation
- Vit K required as cofactor for hepatic synthesis of Prothrombin (II), VII, IX, X, Proteins C and S
- glutamic acid residues of above proteins are carboxylated in reaction involving vitamin K
- allows Ca2+ binding b/c of negative charge
  -> forms complex with Ca2+ allowing it to bind to phospholipids on platelet membrane

Deficiency in neonates
- Hemorrhagic disease of newborn
  -> bleeding various sites including skin, umbilicus and viscera
  -> intracranial bleeding - most serious
- Sterile intestines
  -> no synthesis of vitamin K
  -> routine intramuscular injections of Vit K



Adults
- Caused by
  -> fat malabsorption
  -> prolonged use of broad spectrum antibiotics
- Characterized by
  -> Hematuria
  -> Melena (black tarry stools)
  -> Ecchymoses (bleeding)
  -> Bleeding gums

Effect of Warfarin
- K antagonist
- blocks activity of liver epoxide reductase
  -> prevents regeneration of reduced Vit K
  -> No recycling of Vit K
  -> reduced clotting factors and delayed prothrombin time (INR)



Vitamin D
- D3 in animals (cholecalciferol)
- D2 in Plants (ergocalciferol)

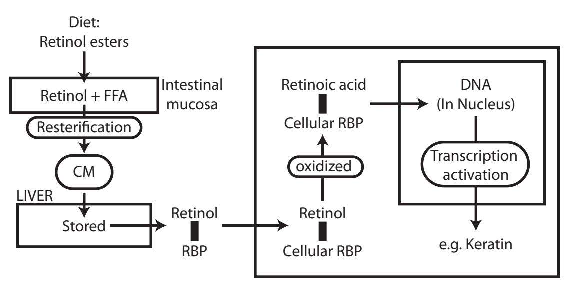
Mechanism of calcitriol
- binds to intracellular receptor proteins
- complex interacts with DNA in nucleus of target cells (intestine)
- selectively stimulate or repress gene expression (similar to steroids)

Actions: (Increase plasma calcium)
- Intestine
  -> stimulates absorption of Ca++ and Phosphate via synthesis of Ca++ binding protein
- Bone
  -> mobilize Ca++ and Phos from bone
  -> Parathormone required
- Kidney
  -> inhibit Ca++ excretion
  -> parathyroid dependent

Deficiency
- Causes
  -> nutritional
  -> inadequate exposure to sunlight
  -> Chronic renal disease (Renal Rickes), chronic liver disease (decreased vit D hydroxylation

Rickets:
- Vit D def in children
- decreased Ca++ absorb from diet
  -> increased parathyroid hormone
  -> increased demineralization of bone
- soft pliable bones
- bow-leg deformity
- Pigeon chest deformity
- frontal bossing

Osteomalacia
- Vit D def adults
- Bones demineralized
  -> susceptible to fracture
- Can be secondary to dietary, renal disease or liver disease



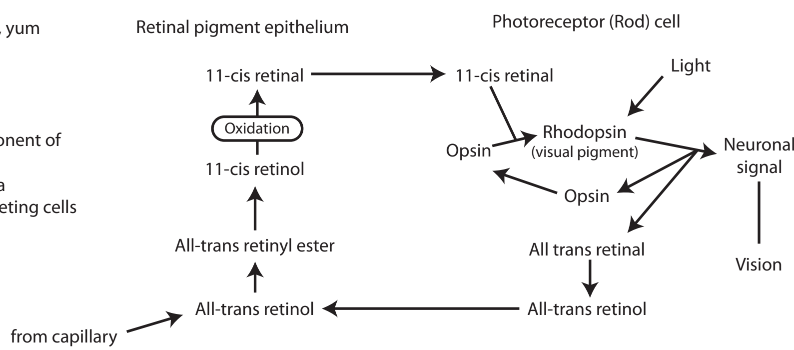
Vitamin A:
- Collectively called retinoids
- Retinol - transport and storage
- 11-cis retinal - required for vision
- Retinol and Retinal
  -> can be easily interconverted
- Retinoic acid
  -> epithelial growth and differentiation
  -> steroid hormone like effects
  -> irreversible

Sources:
- liver, kidney, egg, cream, yum
- Yellow veggies and fruit

Fucntion:
- Vision
  -> 11-cis retinal is component of rhodopsin
- Maintenance of epithelia
  -> especially mucus secreting cells (retinoic acid)
- Growth(retinoic acid)
- Reproduction (retinol)

Absorption and transport of Vit A
1) Diet contains retinol esters
2) hydrolysis by intestinal mucosa releasing retinol and FFA
3) Re-esterification and secretions in CMs taken up by liver and and stored
4) Plasma Retinol binding protein (liver) transports to extra hepatic tissues
5) Tissue contain cellular retinol binding protein carrying retinol into cell

Mechanism of action of retinoic acid in epithelial cells
- similar to steroid
1) Retinol enters target cell
2) oxidized to retinoic acid in cytosol
3) retinoic acid enters nucleus with help from cellular retinol binding protein
4) Retinoic acid binds to nuclear receptors forming activated receptor complex
5) binds to chromatin activating transcription (i.e. Keratin)



Vitamin A deficiency
- most commonly dietary (fat free diets)
- Malabsorption of fats

Signs and symptoms
- Night blindness (earliest symptom
- Xeropthalmia - dry cornea
- Bitot's spots - increased keritinization
- Keratomalcia - corneal erosion and ulceration
- Increased risk of pulmonary infections
- Immune deficiency - weakened innate immunity
- Also less fertility, slowed growth and skin problems

Treatment
- Retinoic acid is used to treat:
  -> severe acne
  -> psoriasis
- All-trans retinoic acid
  -> treatment of acute promyelocytic leukemia

Toxicity
- Must be careful during pregnancy
  -> may be teratogenic (cong malf)
  -> spontaneous abortions
- Hypervitaminosis A
  -> raised intracranial pressure
  -> headaches mimicing brain tumors

Vitamin E
- α–TOCOPHEROL most active form
- Anti-oxidant (lipid soluble)
  -> prevents peroxidation of lipids
  -> Scavanges free radicals generated
- normal diet adequate (only seen with fat malabsorption

Vit E and Vit C work together
- Vit E
  -> Works in cell membranes
- Vit C
  -> Works in cytosol or cell organelles

Deficiency:
- Hemolytic anemia
  -> abnormal cell membrane
- Reduced deep tendon reflexes
  -> b/c axonal degeneration

Vitamin C (Ascorbic Acid)
- Water soluble
- collagen synthesis and wound healing
  -> cross linking collagen
  -> coenzyme for hydroxylation of proline and lysine residues in collagen
- water soluble anti-oxidant
  -> scavenger of free radicals
  -> regenerates anti-oxidant form of Vitamin E

Deficiency
-Scurvy
  -> Perifollicular hemorrhages (fragile blood vessels)
  -> Sore, spongy gums (bleeding)
  -> Loose teeth
  -> Bleeding into joints
  -> frequenct bruising
  -> impaired wound healing
  -> similar to K but different

Vitamins
- Water-soluble
  - Non-B-complex
    - C: Ascorbic acid
  - B-complex
    - Energy releasing:
      - B1: Thiamine
      - B2: Riboflavin
      - B3: Niacin
      - Biotin
      - Pantothenic acid
    - Hematopoietic (RBC/neural development)
      - Folic acid
      - B12
    - Other
      - B6: pyridoxine
- Fat-soluble
  - A: Retinol, beta carotenes
  - D: cholecalciferol
  - K: phylloquinones, menaquinones
  - E: locopherols

Vitamin B Complex
- Thiomine
- Riboflavin
- Niacin
- Pyridoxine
- Biotin
- Pantothenic acid (CoA)
- Cobalamin
- Folic Acid

Water soluble vitamins usually NOT stored-
Daily supplements essential
-- Exception! B12 stored in liver

Vitamin B1 (Thiamine)
- TPP - coenzyme form
- Functions
  -> Oxidative decarboxylation of alpha keto acids
  -> Helps maintain neural membranes and normal nerve conduction
  -> PDH, α-Keto Acid DH, BC(alpha-keto)A DH
  -> Coenzyme for transketolase in Pentose phosphate shunt
- B1 supplemenetation for Maple syrup urine disease
- refined foods:
  -> polished rice, white flour, white sugar
  -> are defficienct

Thiamine deficiency

BeriBeri:
- affects highly aerobic tissue (Brain, cardiac muscle)
- Polyneuropathy
  -> disruption of motor, sensory and reflex arcs
  -> Progress to Dry Beri Beri (paralysis)
- Cardiovascular symptoms
  -> Web Beri Beri

Pedal

Werneke-Korsakoff syndrome:
- Associated with chronic Alcoholism
- Ophthalmoplegia and nystagmus
  -> to and fro movement of eyeballs
- Ataxia, confusion, disorientation and memmory loss
- Confabulation (telling stories)
- Diagnosis
  -> incrase in erythrocyte transketolase activity on TPP addition

Vitamin B2 (Riboflavin)
- Coenzyme forms
  -> FMN (flavin mononucleotide)
  -> FAD (flavin dinucleotide)
- Participates in oxidation
  1) TCA cycle + complex 1 (ETC)
  2) beta oxidation (Succinate DH, PDH, (Fatty) Acyl CoA DH...etc)

MCAD

Riboflavin deficiency
- Nutritional signs and symptoms
  1) cheilosis - areas of pallor, cracks and fissures at angles of mouth
  2) Glossitis - inflammation and atrophy of tongue
  3) facial dermatitis

Biotin (biocytin)
- Prosthetic group for most carboxylation reactions
- Enzymes requiring biotin:
  1) Pyruvate carboxylase - (gluconeogesis)
  2) Acetyl-CoA carboxylase - (FA synth)
  3) Propionyl-CoA carboxylase - (Odd chain)

Deficiency
- Inherited deficiency leads to multiple carboxylase deficiency
- Avidin
  -> present in raw egg white
  -> inhibits biotin

Treatment:
- Biotin supplementation improves symptoms in children with mult. carboxylase deficiency

Vitamin B3 (Niacin)
- Coenzyme forms
  -> NAD+
  -> NADP+
- Coenzymes in oxidation-reduction reactions
  1) NAD - Dehydrogenases
  2) NADP - reactions in HMP shunt and fatty acid synthesis

Therapeutic uses:
- Treatment of type IIb hyperlipoproteinemia
  -> inhibits lipolysis in adipose tissue
  -> reduces productino of FFA

Niacin deficiency

Pellagra:
- Chracterized by 3 D's
  1) Dermatitis- necklace like skin damage
  2) Diarrhea
  3) Dementia
  4) Death (if not treated)
- Can be caused by corn based diet (corn deficienct ion niacin and tryptophan)
- Patients with Hartnup's disease can have pellagra-like symptoms
  -> defect in tryptophan absorption
- Patients with carcinoid syndrome may also have pellagra

Treatment:
- Tryptophan
  -> used to synthesize NAD+ and NADP+

Vitamin B6 (pyridoxine)
- AKA pyridoxine, pyridoxal, pyridoxamine
- PLP (pyridoxal phosphate) precursor
- Co-enzyme for
  1) Transamination (amino acid metabolism)
  2) Amino acid decarboxylation (synthesis of neurotransmitters)
  3) Condesation (ALA synthase)
  4) Conversion of homocysteine to cysteine

Deficiency

Isoniazid (anti-TB drug)
- inactivates Pyridoxine
- Supplements given with drug

Signs and symptoms
- Microcytic anemia (ALA synthase)
- Peripheral neuropathy
- Increased risk of cardiovasc disease
  -> due to high plasma homocystine
- Seizures

Treatment:
- Children with homocystinuria respond to dietary B6 supplements

Nutritional anemias:

Microcytic (MCV < 80)
- Due to reduced synthesis of heme
- Also seen in lead poisoning
1) Deficiency in iron
2) Deficiency in copper
3) Deficiency in pyridoxine

Normocytic (MCV 80-100)
1) protein calorie malnutrition

Macrocytic (MCV > 100)
- reduced cell division
1) Deficiency in B12
2) Deficiency in folate

Vitamin B12: 2 Major reactions
1) Homocysteine methyltransferase
- Synthesis of methionine
- Also converts methyl tetrahydrofolate to THF (Active form required for DNA synthesis)
2) Methylmalonyl CoA mutase
- Odd number chain FA

Deficiency:
Folate trap
- methyltetrahydrofolate (not THF)
- results in macrocytic anemia

Methylmalonic aciduria
- Neruological manifestations
- Methylmalonate levels high in circulation
- Dietary supplementation is useful with inherited disorder

Pernicious anemia
- Lack of intrinsic factor from gastric parietal cells
- Most common cause is improper absorption
- Macrycytic anemia
  -> Macrocytes in peripheral blood
  -> Megaloblasts in bone marrow
    - Megaloblastic anemia
  -> Due to 2nd deficiency of folate
- Neuropsychiatric symptoms
  -> Myelin degeneration both motor and sensory
  -> Due to methylmalonyl accumulation
  -> May be present in absence of anemia

Folic Acid
- One Carbon Metabolism
- important in purine and pyrimidine synthesis
  -> e.g. thymidylate (DNA synthesis)
- Most cells receive folate as methyl THF
  -> conversion to THF requires B12

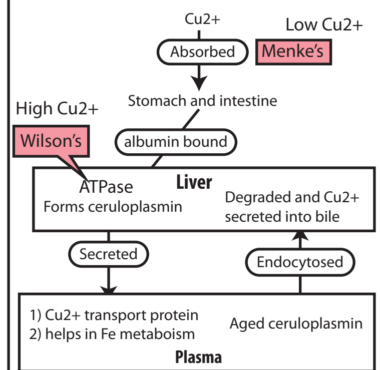
Deficiency:
- Dietary inadequacy (green leafs)
- impaired absorption
- increased requirment in pregnancy
- Methotrexate (Folate antagonist)
- B12 deficiency (Folate trapping)

Megaloblastic anemia
- Macrocytes in peripheral blood smears
- Megaloblasts in bone marrow
- Due to
  -> deminished synthesis of purines and pyrimidines
  -> Inability of cells to synthesize DNA
  -> Delayed mitosis; larger cells

Treatment:
- Megaloblastic anemia
  -> never treat with only THF
  -> include B12 until cause established
- Pregnancy
  -> Increased requirement
  -> prevent neural tube defects
  -> recommended at time of conception
  -> and during first trimester

Microcytic anemia - Cu2+, Fe2+, Lead, PLP
Macrocytic - B12, Folate
Normocytic - protein/calorie malnutrition

Copper: - Important co-factor in redox reactions
1) Cytochrome C
  -> Complex IV of ETC
2) Superoxide dismutase
  -> Detox free radical oxidase
3) Lysyl oxidase
  -> Synthesis of collagen
4) Tyrosinase
  -> Melanin synthesis
5) Dopamine β-hydroxylase
  -> Neurotransmitter synthesis
- Cu2+ forms ceruloplasmin in liver
  -> copper transport protein in liver
  -> helps in iron metabolism
Copper metabolism:



Cu2+ Deficiency: Nutritional
Signs and symptoms
1) Microcytic anemia
  -> smaller RBCs
  -> ceruloplasmin (ferroxidase) required for iron metabolism
2) Degradation of vascular tissue
  -> decreased lysyl oxidase activity
3) Defects in hair

Menke's syndrome (Menke's kinky hair syndrome)
- Inherited (X-linked) defect in copper absorption from GI tract
- Low plasma and tissue copper
- Hair twisty grayish and kinky
- Leads to aneurysms and cerebral dysfunction

Wilson's Disease ***
- Autosomal recessive
- copper metabolism disorder
- toxic levels of Cu2+ in liver brain and eye
- Chromosome 13 (30 mutations identified)
- Defect in Cu2+ transporting ATPase in liver
- protein needed to attach Cu2+ to ceruloplasmin
  -> also excrete Cu2+ into bile
- accumulation in liver spills into plasma and other tissue
1) RBC damage - hemolysis
2) KF rings (eye) - deposition into cornea)
3) Brain - neuro damage - causing neuropsychiatric symptoms (basal ganglia
Lab diagnosis
1) decreased serum ceruloplasmin
2) increased urinary excretion of copper
3) increased hepatic copper content

Iron
- Important for heme synthesis and redox reactions
- Absorbed into mucousal cells after conversion of ferric to ferrous (absorbable)
- Converge happens in stomach because:
  1) low pH
  2) Vitamin C
- Heme iron easier in meat than inorganic Fe (plants)
- Absoprtion TIGHTLY regulated
  -> High body iron stores, less iron absorbed

Iron transport and storage
- Ceruloplasmin (ferroxidase)
  -> release of ferrous iron from intestinal cells
  -> forms ferric iron - needed for blood transport
- Transferrin
  -> transport protein for ferric ion in blood plasma
- Iron stored as ferritin and hemosiderin (ferric) in
  1) liver
  2) RES

Iron Deficiency:
- Most common nutritional deficiency
- At risk: infants, pregnant moms and blood doners
- Chronic bleeding is underlying factor

Clinical features:
1) Hypochromic microcytic - Iron deficiency anemia
2) Fatigue and pallor
3) Weakness
4) Brittle nails
5) Pica - appetite for soil

Hereditory Hemochromatosis
- Autosomal recessive - Most common Genetic disorder
- excessive absorption of iron
- Accumulation parenchymal organs, most importantly:
  1) Liver
  2) Pancrease
    -> Bronze diabetes (Skin Fe, low insulin b/c iron disrupts Beta cells in pancrease)
- HFE gene
  -> on basolateral surface of intestinal crypt
  -> senses iron balance
  -> defect causes loss of sensing = high Fe absorbtion

Tissue damage
- Lipid peroxidation through free radicals
- DNA damage

Clinical features:
1) More common in males (age 40)
2) Liver damage - cirrhosis; hepatocellular carcinoma
3) Diabetes - destruction of pancrease
4) Cardiac dysfunction
5) Acute synovitis
6) Brownish skin pigmentation (Bronze color of skin)